

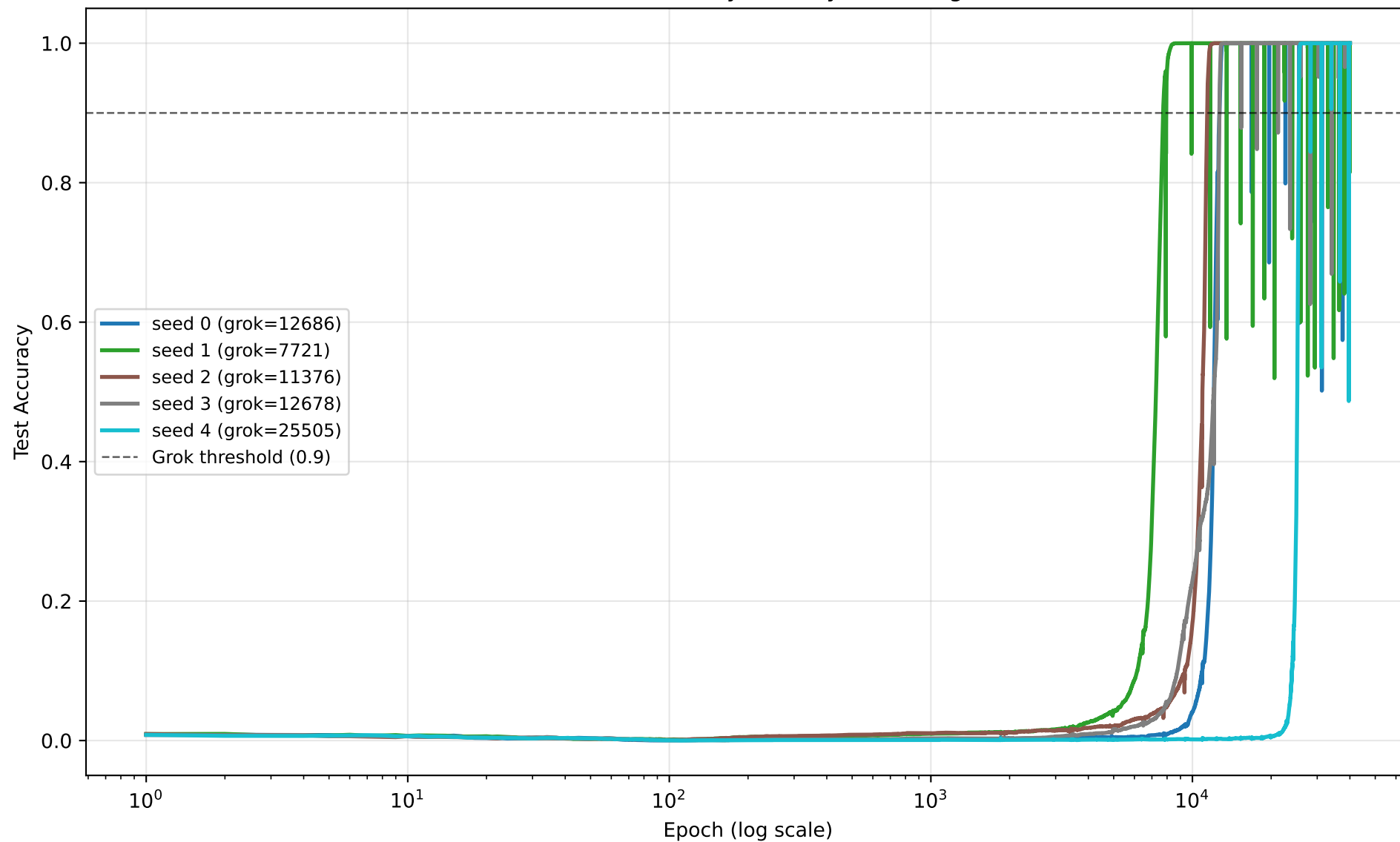
NANDA-UNIFIED BENCHMARK — RESULTS REPORT

results_dir : /Users/jonathanjohn/Documents/grokking-benchmark/results/nanda_unified
seeds : [0, 1, 2, 3, 4]

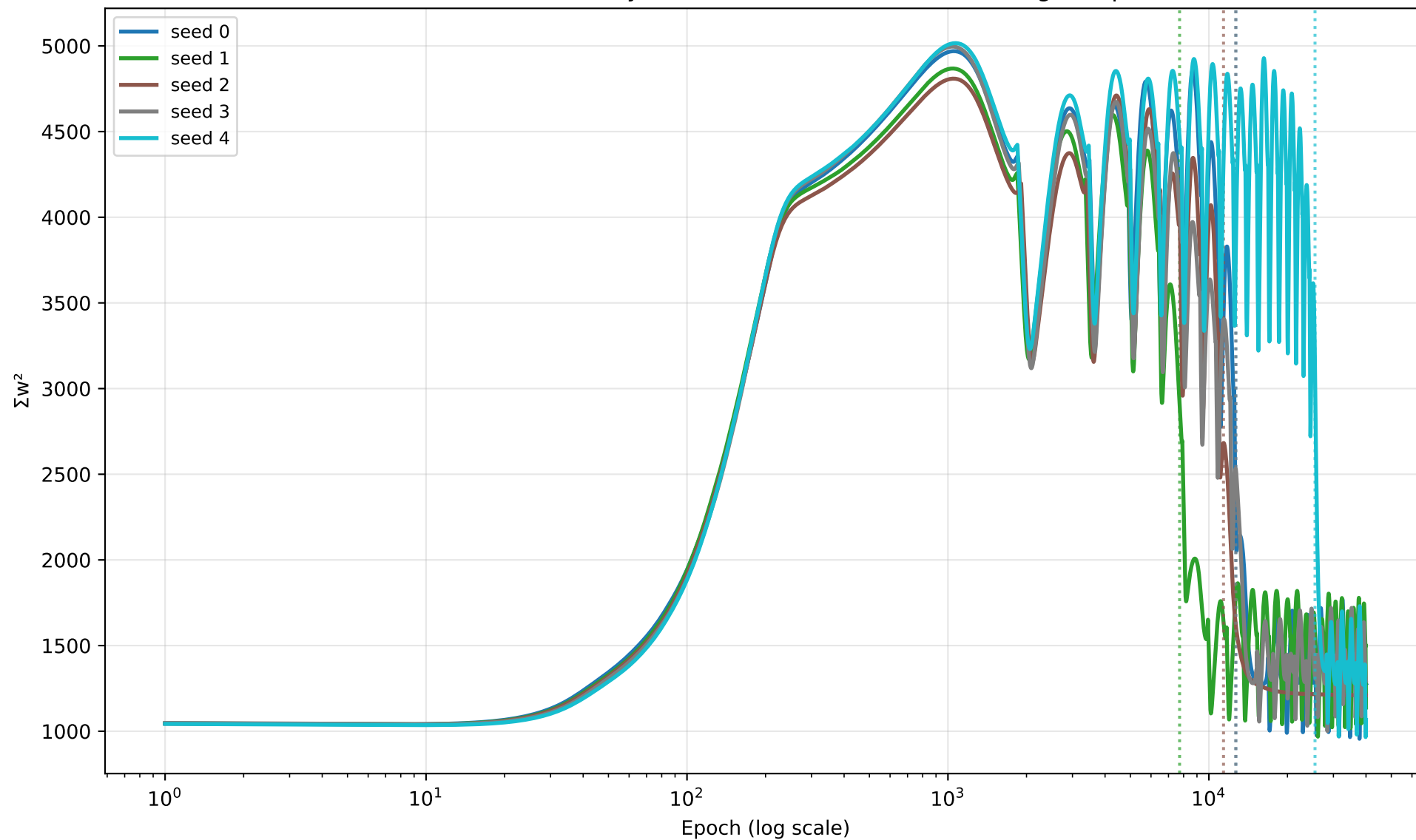
=== AGGREGATE SUMMARY ===

n_seeds : 5
epochs : 40000
grok_epoch_mean : 13993.2
grok_epoch_std : 6035.731783305153
grok_epochs : [12686, 7721, 11376, 12678, 25505]
n_limit_cycle : 0 / 5

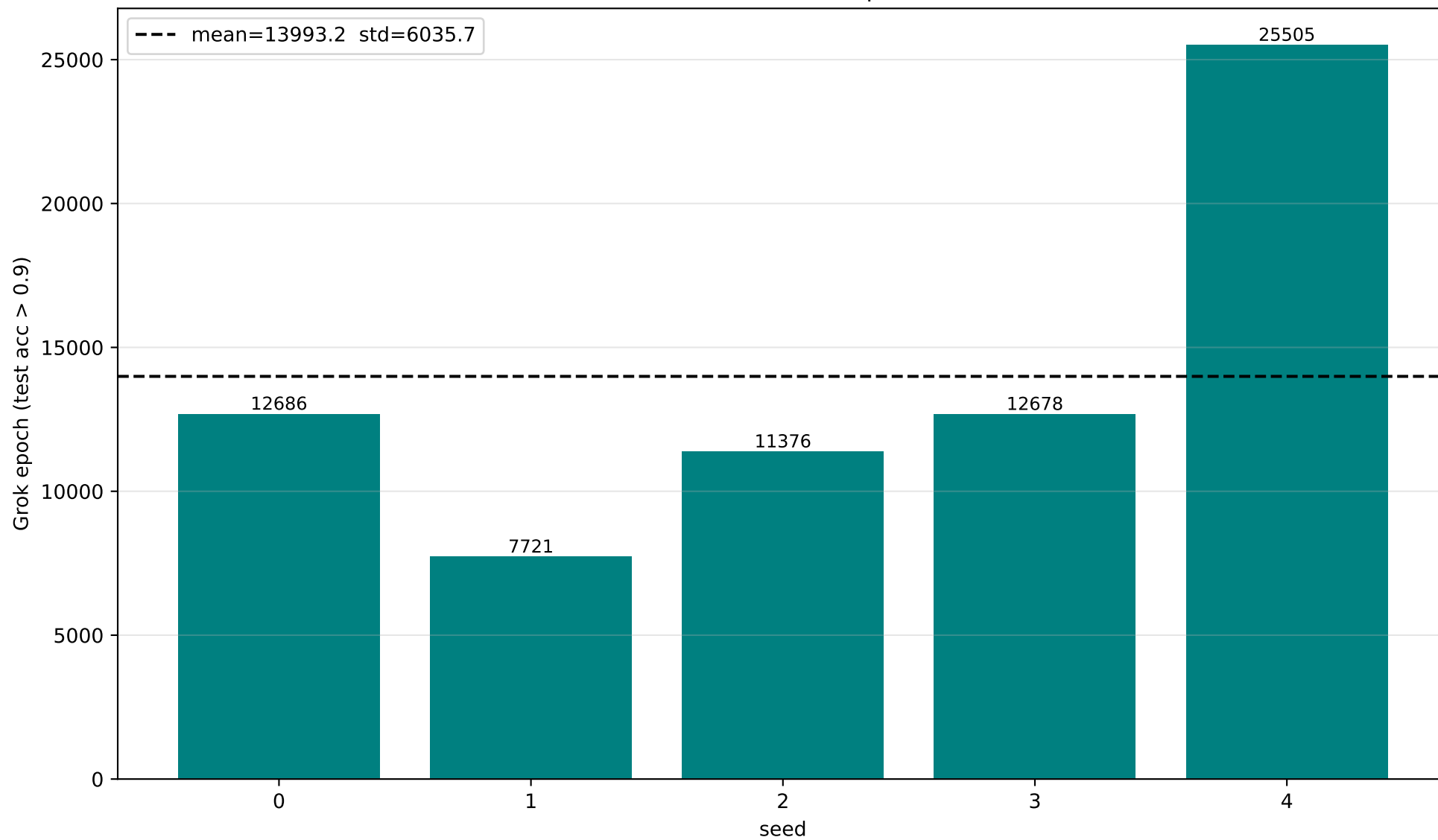
All seeds — Test Accuracy Overlay (Grokking Curves)



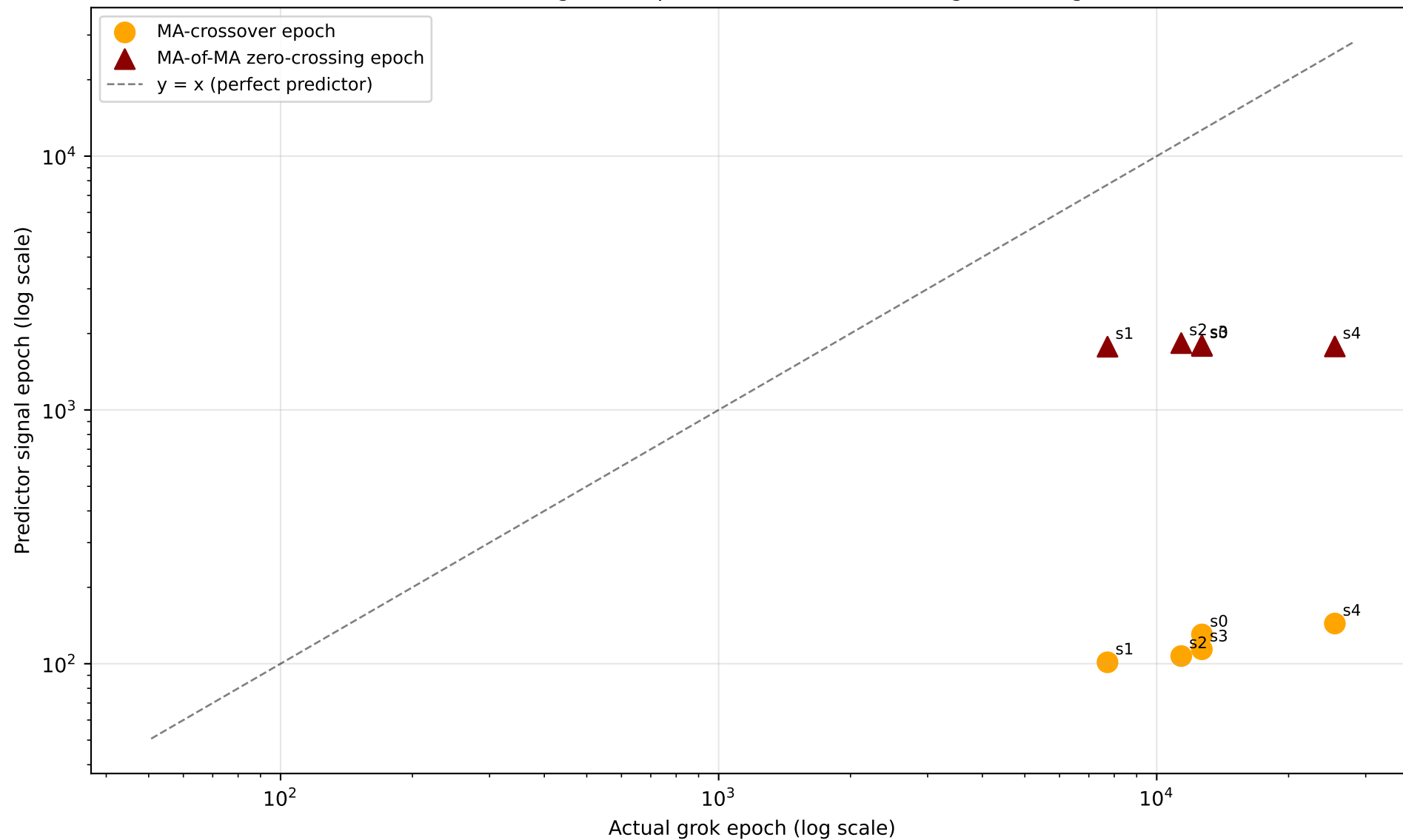
All seeds — Σw^2 Overlay (dotted lines = each seed's own grok epoch)



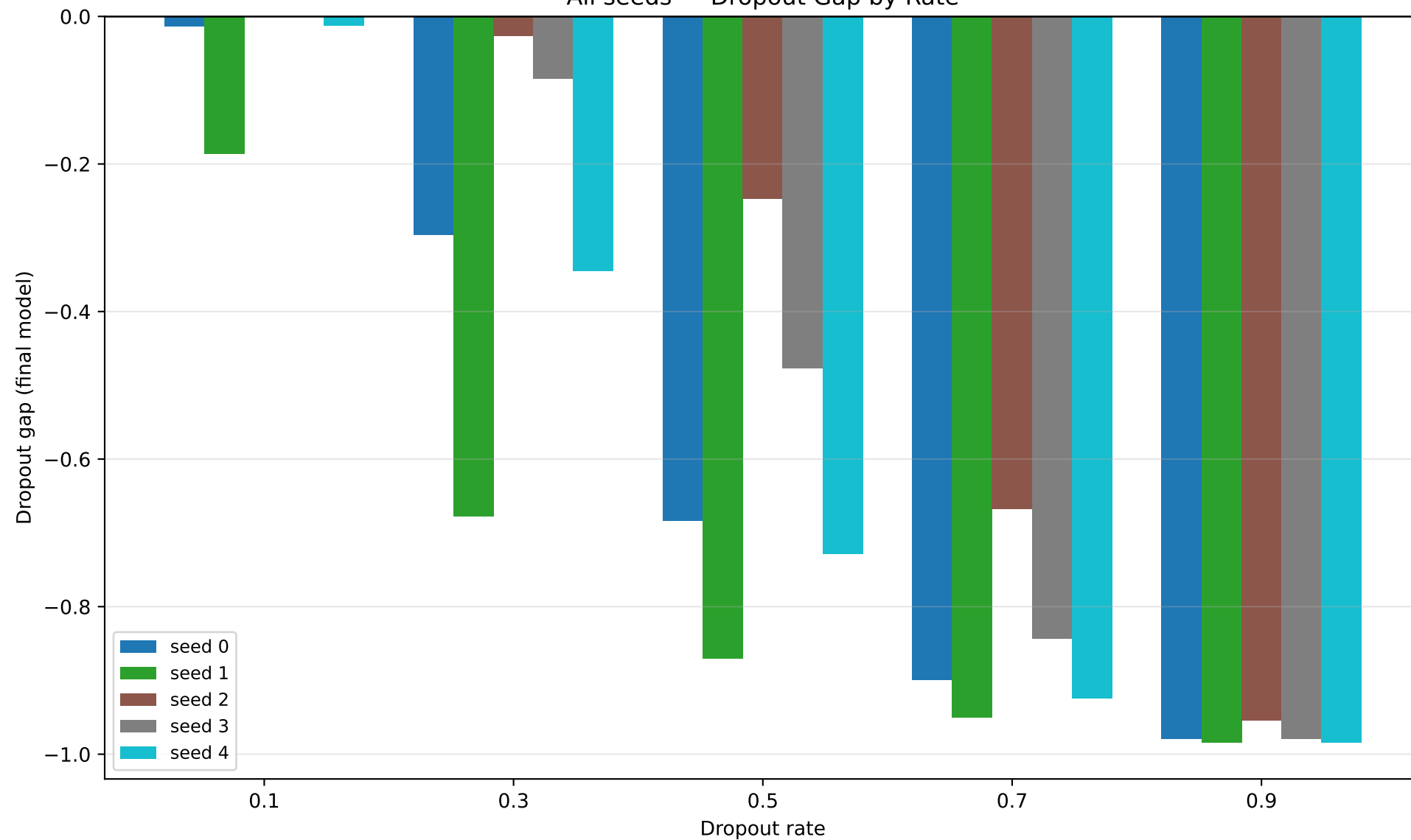
All seeds — Grok Epoch



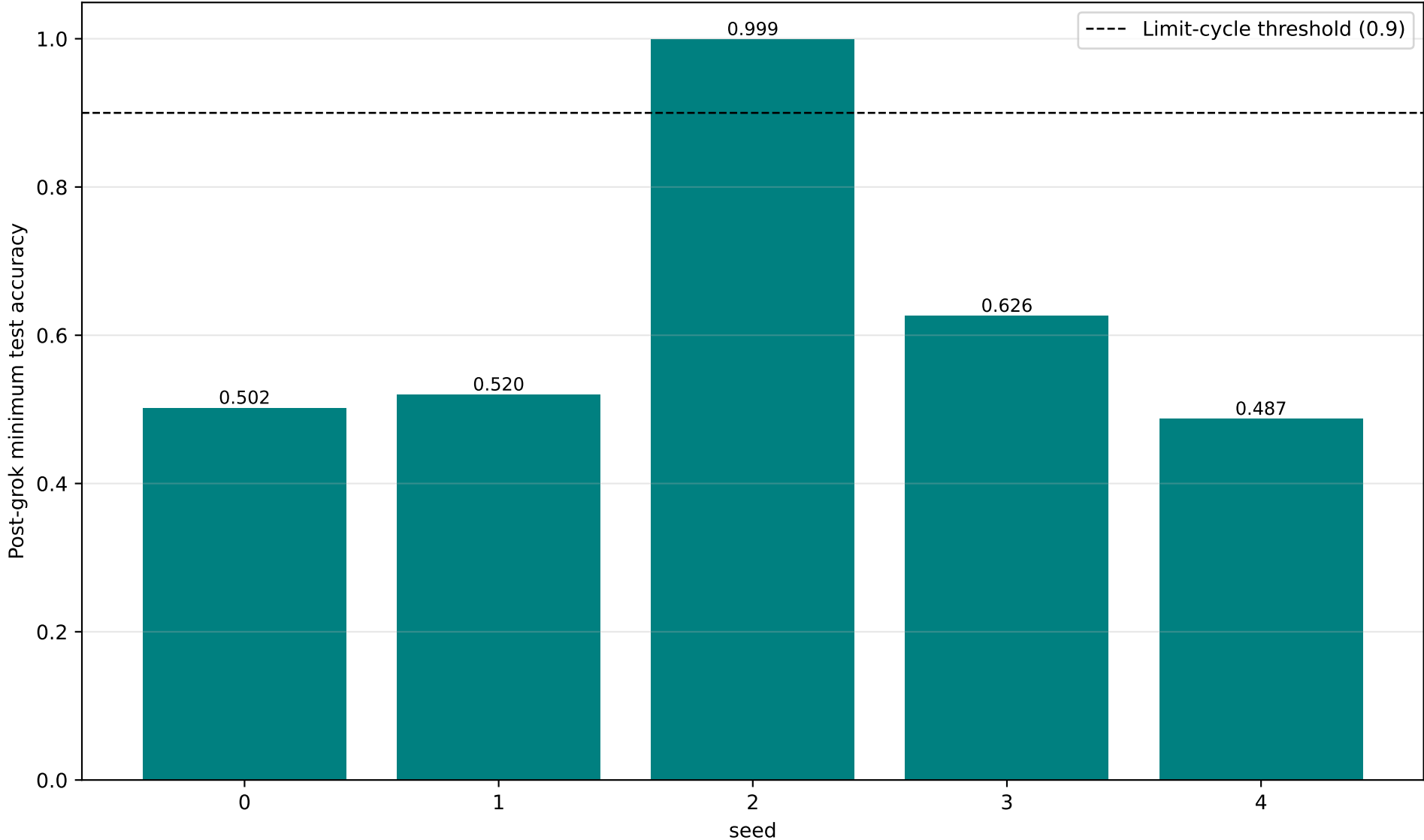
L2-Norm predictor signal vs. actual grok epoch, across seeds
(flat / off-diagonal = predictor does not track grok timing)



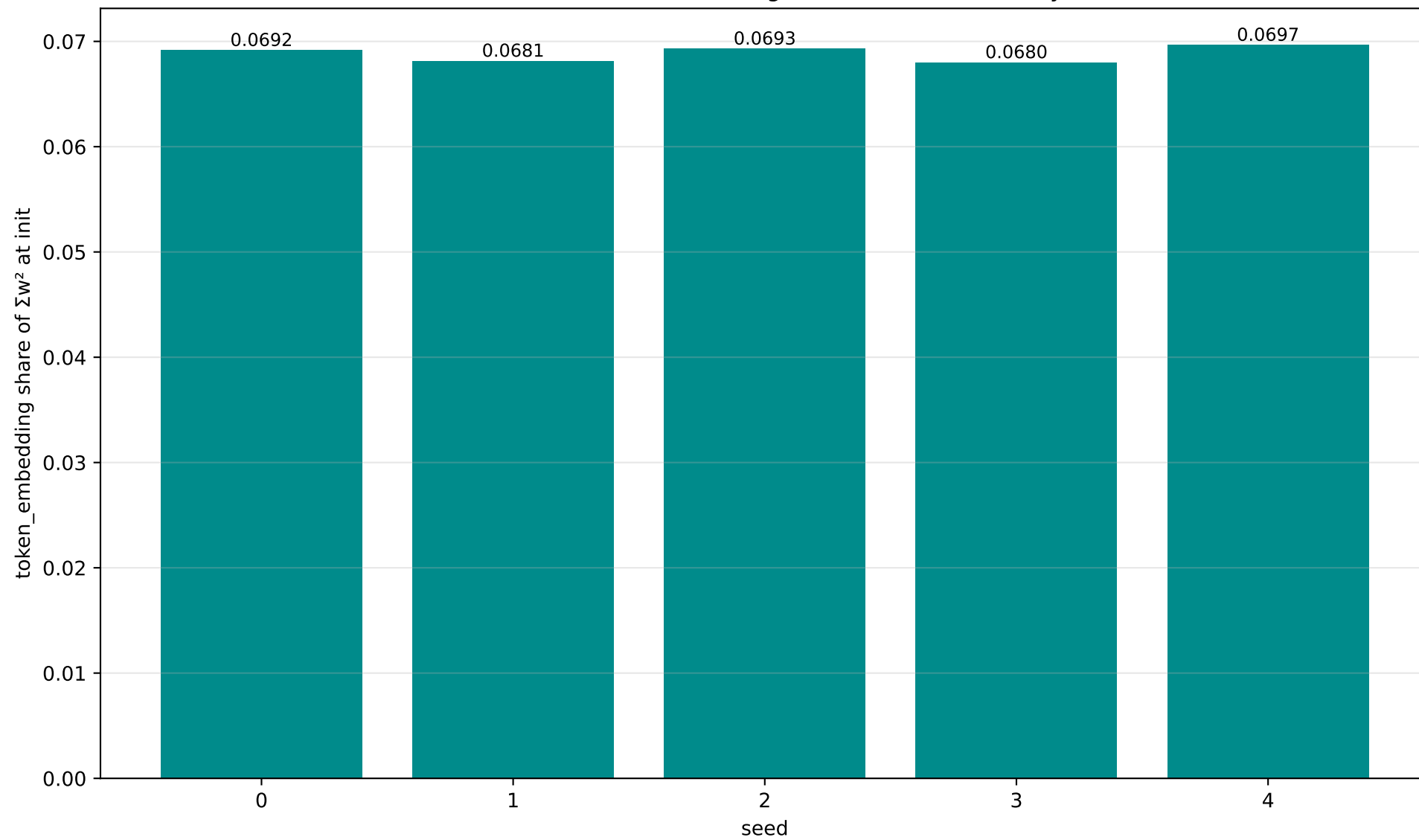
All seeds — Dropout Gap by Rate



All seeds — Post-Grok Limit-Cycle Check (min test acc after settling)



All seeds — Init Token-Embedding Share (small-init sanity check)



seed 0 – full numbers

seed 0 (modulus=113 epochs=40000 wall_time=4918.4s)

grok_epoch : 12686
final_train_acc : 1.0000
final_test_acc : 1.0000
l2_norm init -> final : 32.3335 -> 35.7125
sum_w2 init -> final : 1045.457 -> 1275.381
token_embedding_share (init) : 0.0692

L2 predictor:

MA-crossover epoch : 130.43
MA-of-MA zero-crossing epoch : 1786.54
noise_floor : 0.000471
windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

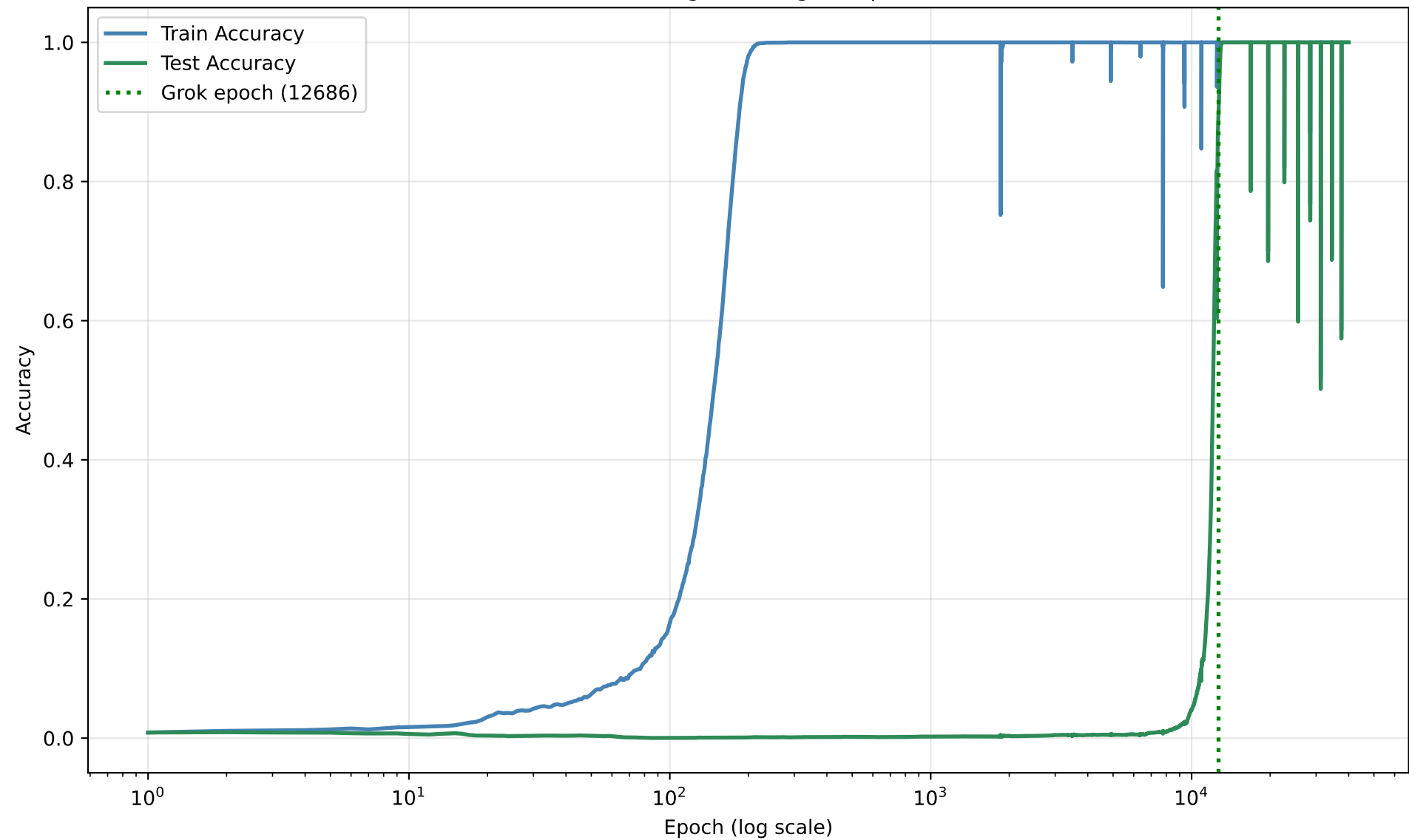
Dropout gap by rate:

p0.1=-0.0135 p0.3=-0.2957 p0.5=-0.6840 p0.7=-0.8989 p0.9=-0.9793

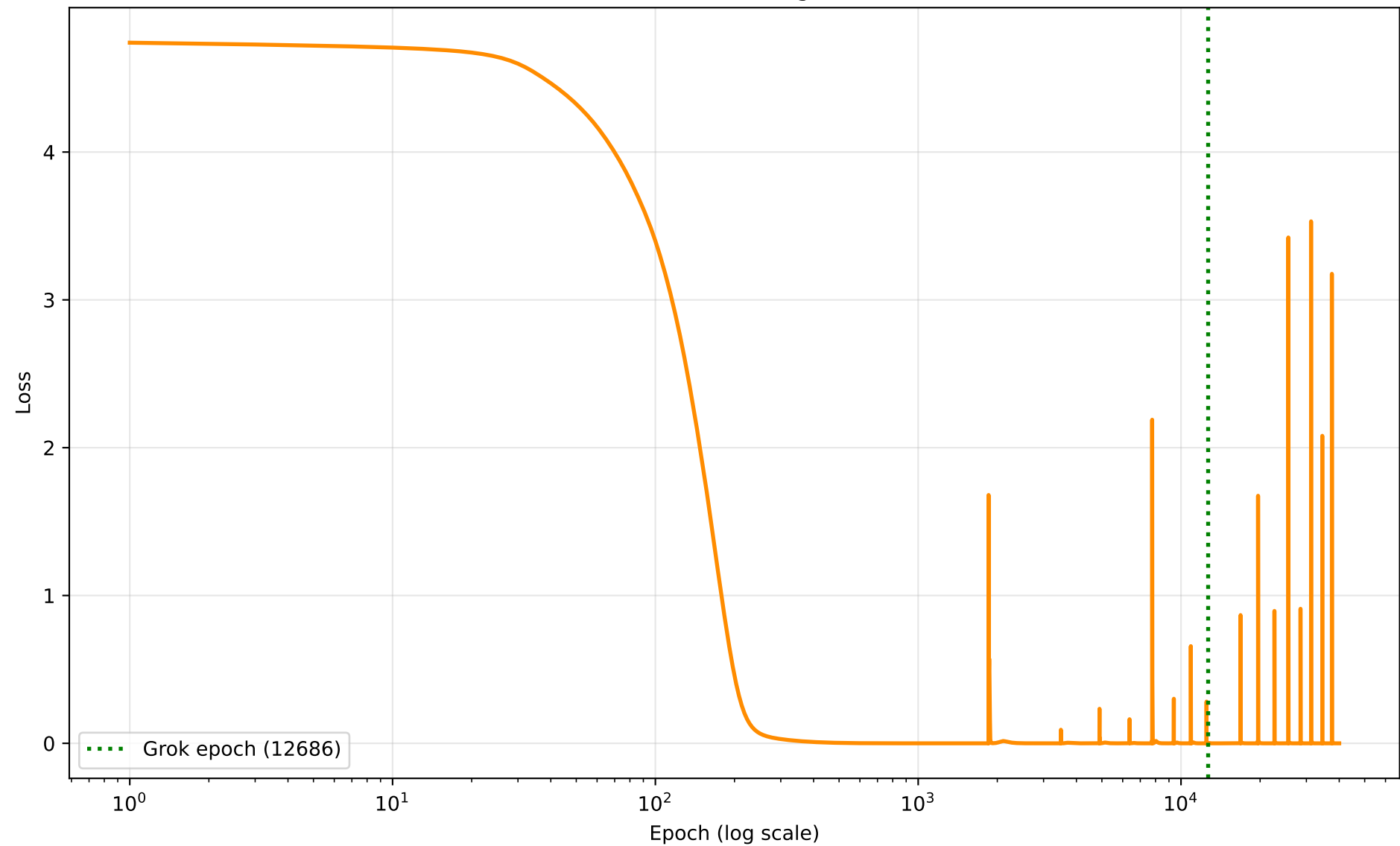
Limit-cycle check : stable

window_start=13186 post_grok_min=0.5017 post_grok_std=0.0089
post_grok_final=1.0000 dips<0.9=35

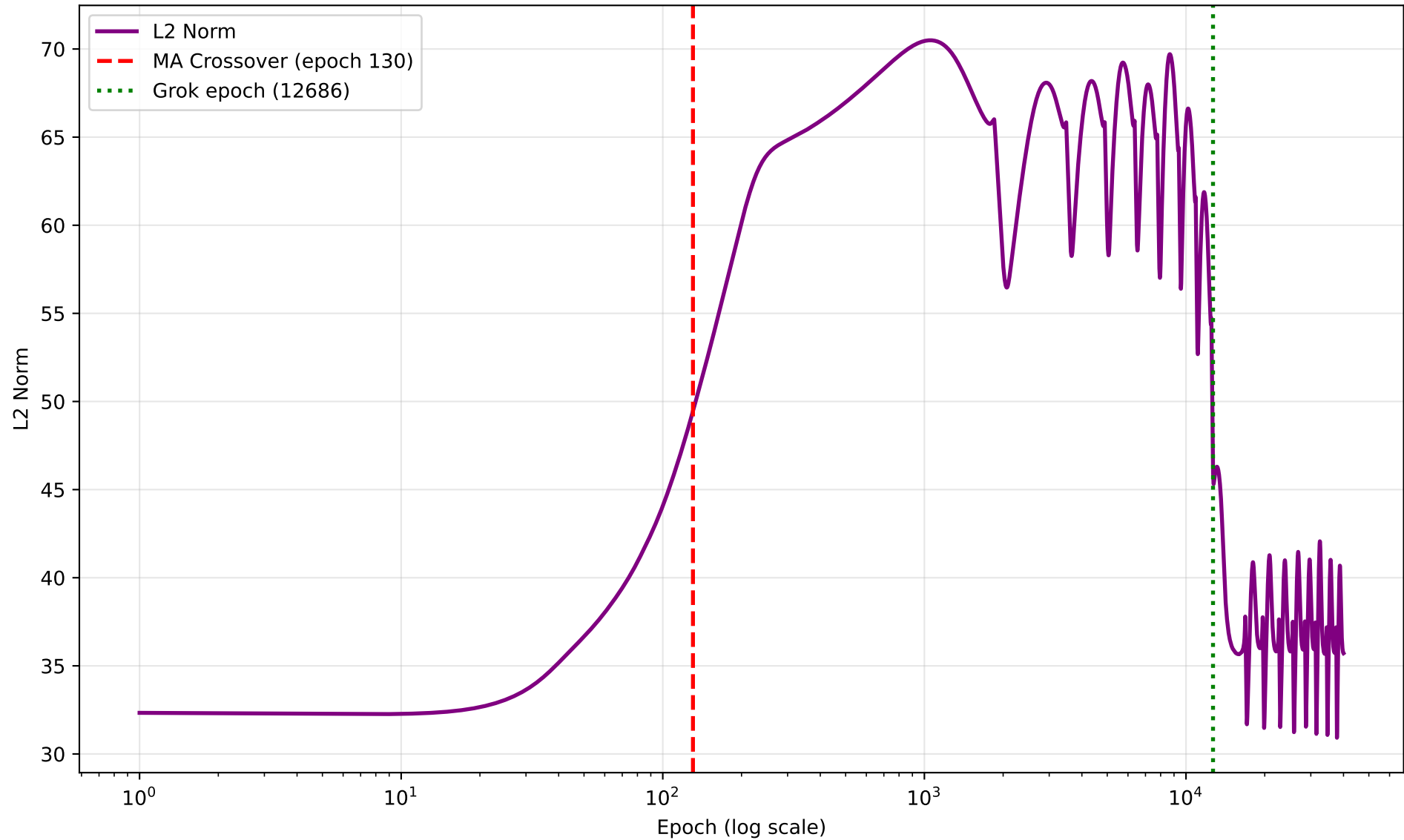
seed 0 — Grokking Curve (grok epoch 12686)



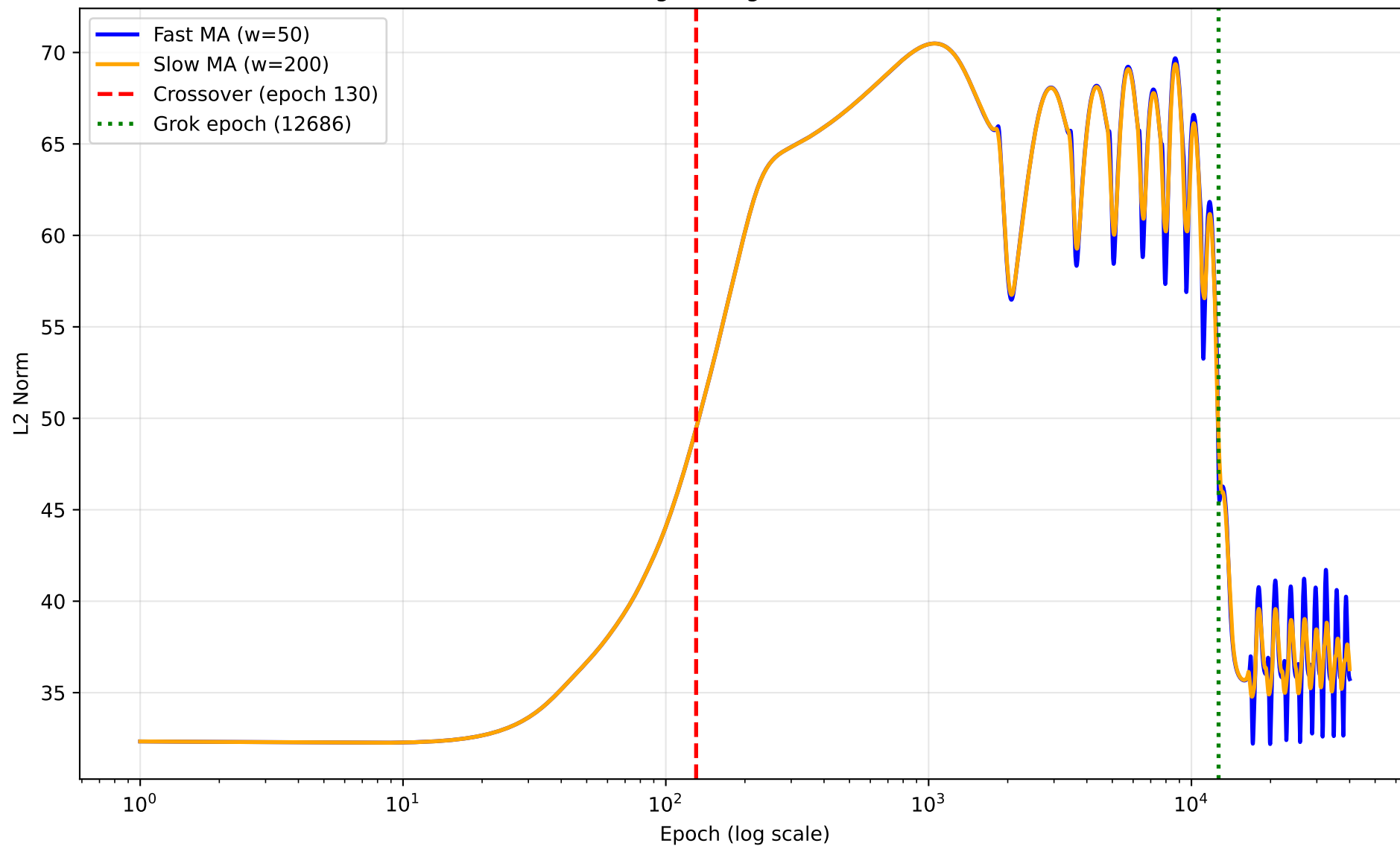
seed 0 — Training Loss



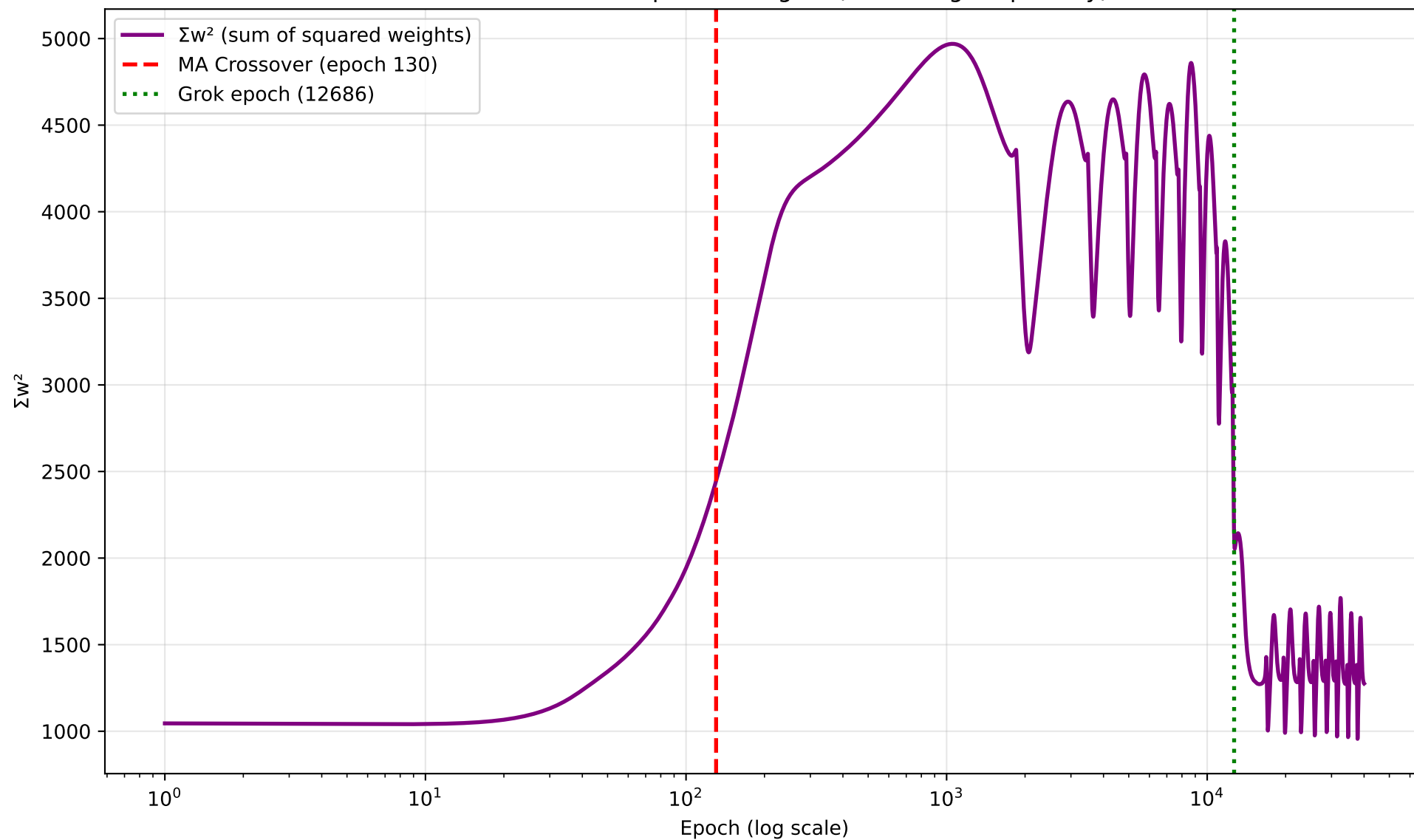
seed 0 — L2 Norm of Model Weights



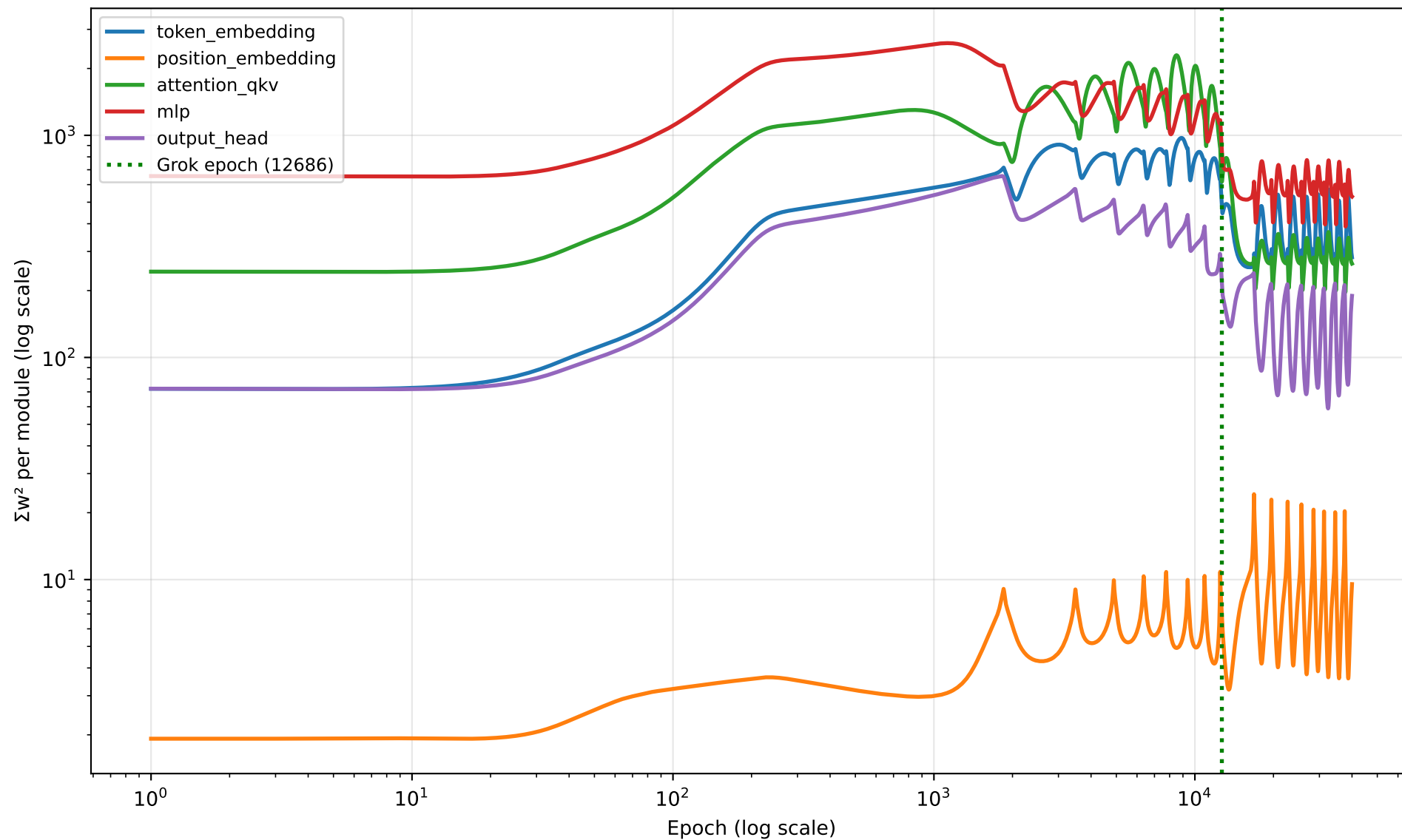
seed 0 — Moving Average Crossover Detection



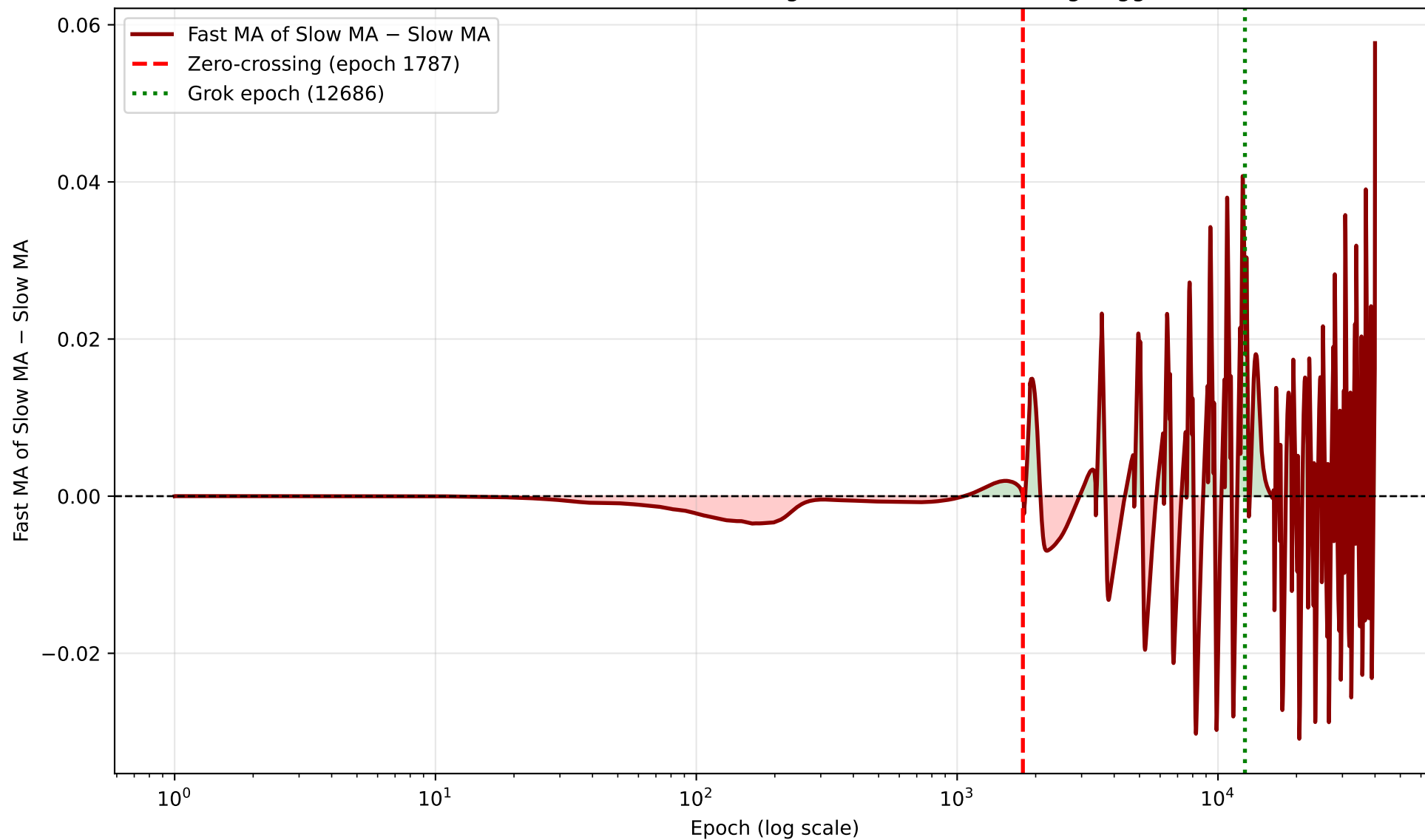
seed 0 — Sum of Squared Weights (Nanda Fig. 7 quantity)



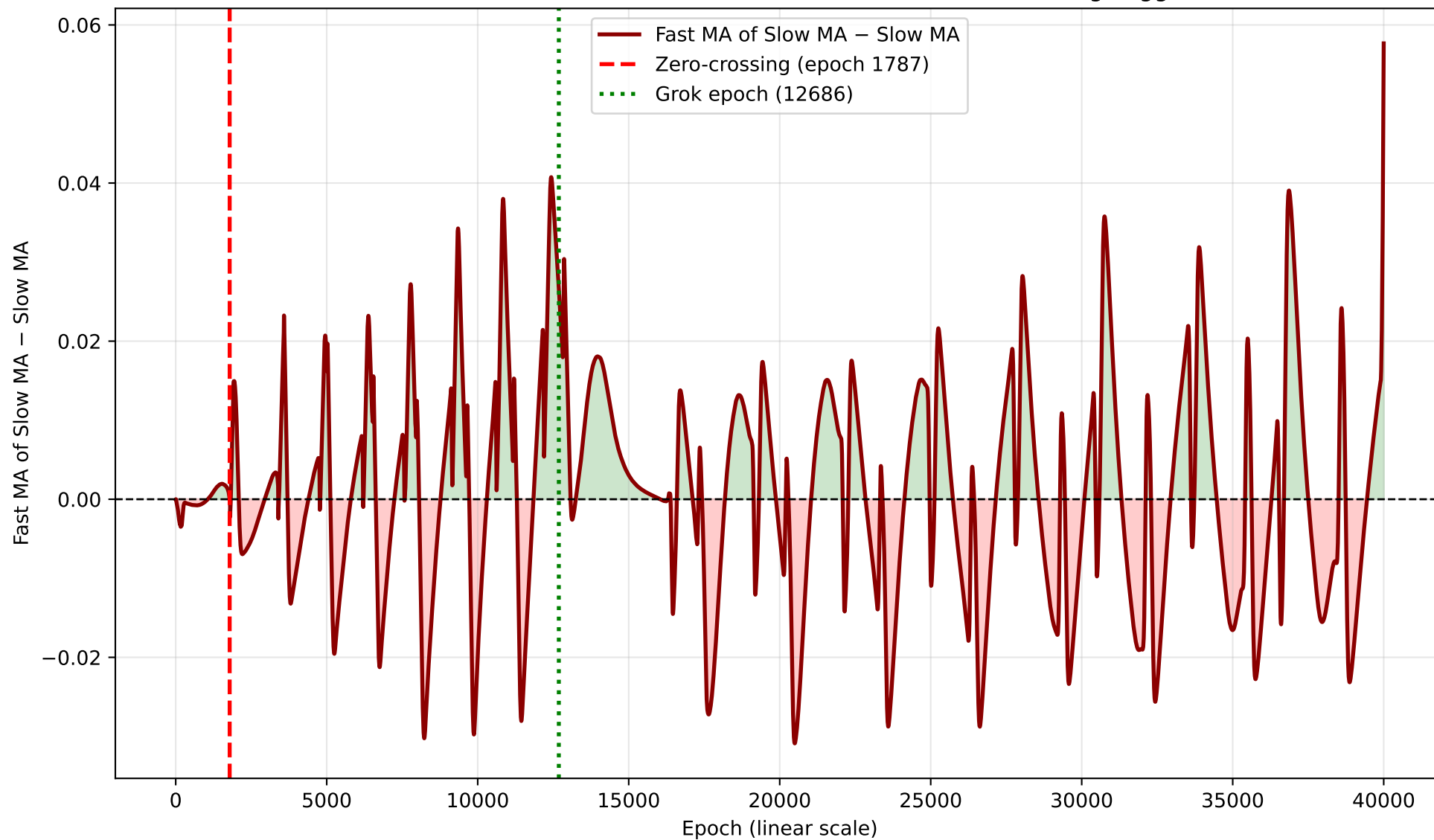
seed 0 — Per-Module Σw^2 Breakdown

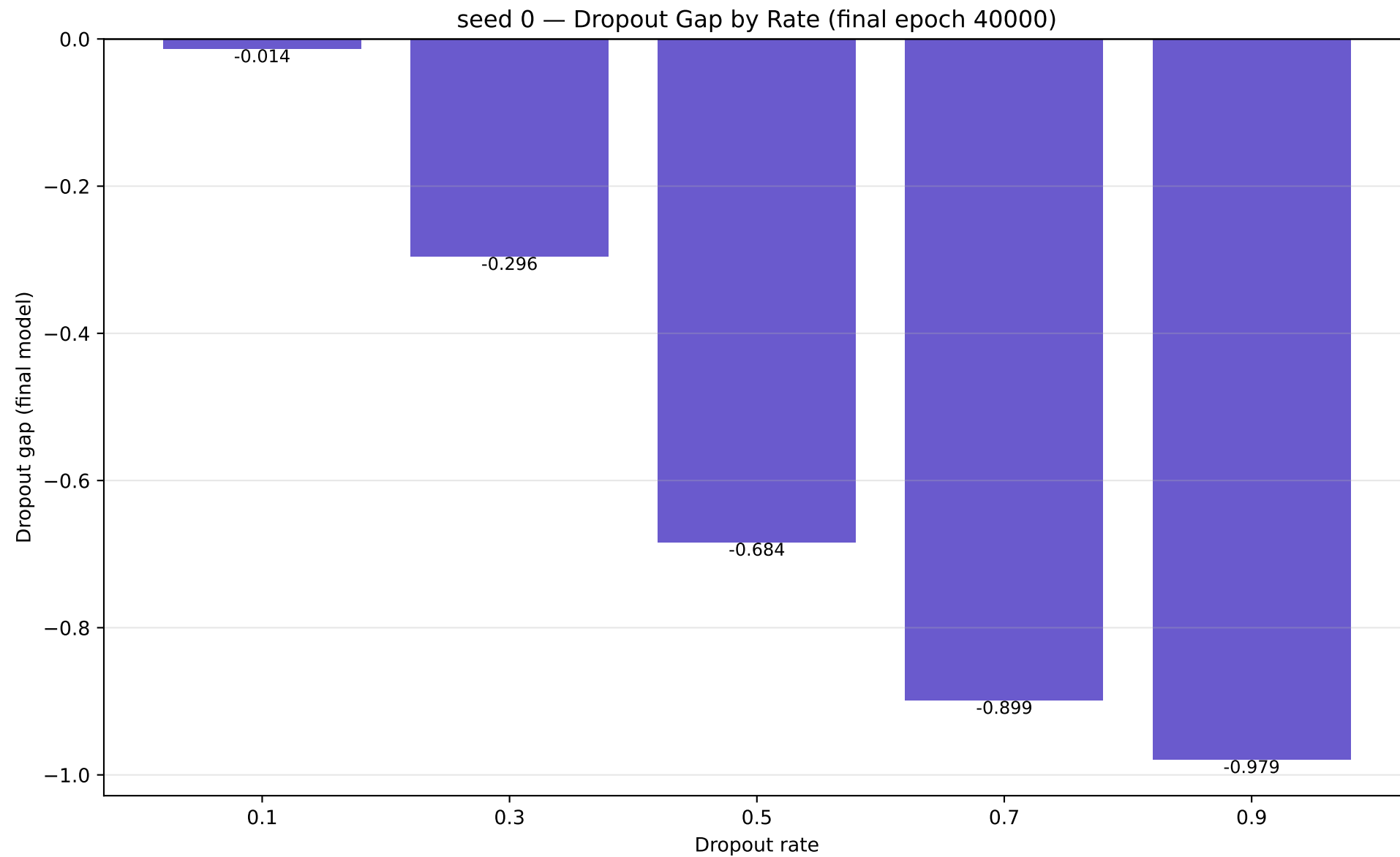


seed 0 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger



seed 0 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger





seed 1 – full numbers

seed 1 (modulus=113 epochs=40000 wall_time=4911.0s)

grok_epoch : 7721
final_train_acc : 1.0000
final_test_acc : 1.0000
l2_norm init -> final : 32.3437 -> 33.7408
sum_w2 init -> final : 1046.116 -> 1138.439
token_embedding_share (init) : 0.0681

L2 predictor:

MA-crossover epoch : 101.30
MA-of-MA zero-crossing epoch : 1774.27
noise_floor : 0.000473
windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

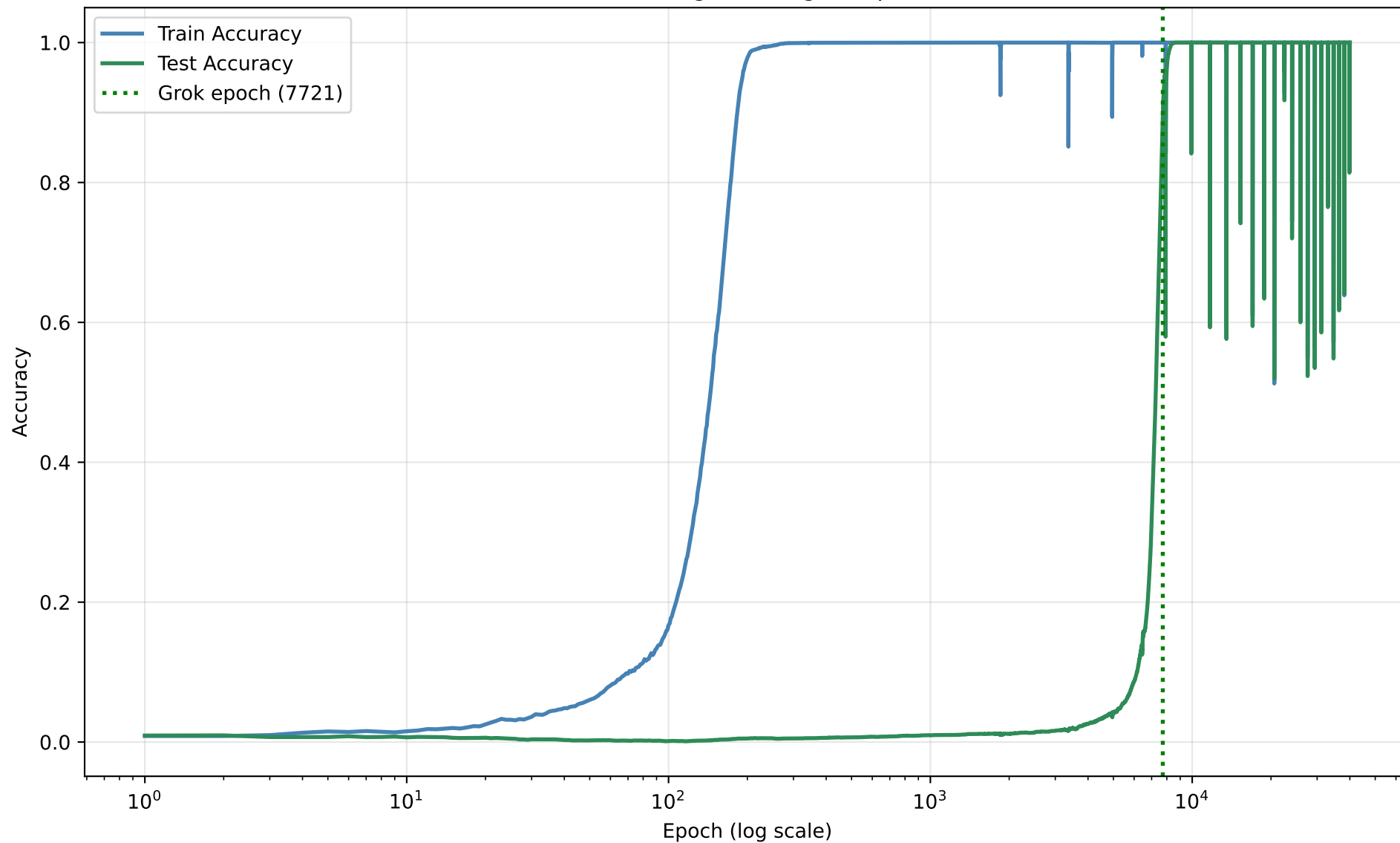
Dropout gap by rate:

p0.1=-0.1865 p0.3=-0.6778 p0.5=-0.8707 p0.7=-0.9503 p0.9=-0.9842

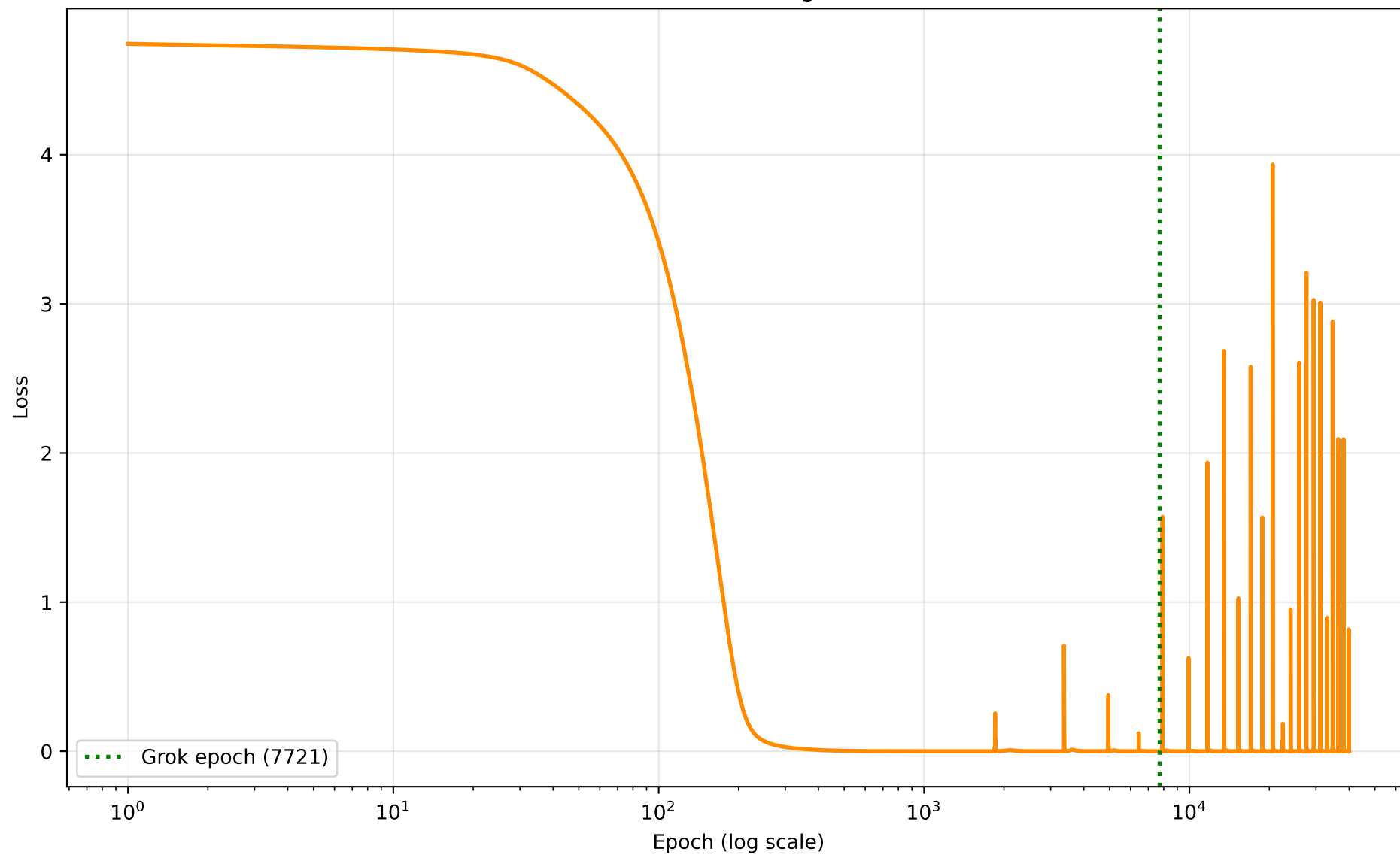
Limit-cycle check : stable

window_start=8221 post_grok_min=0.5196 post_grok_std=0.0136
post_grok_final=1.0000 dips<0.9=83

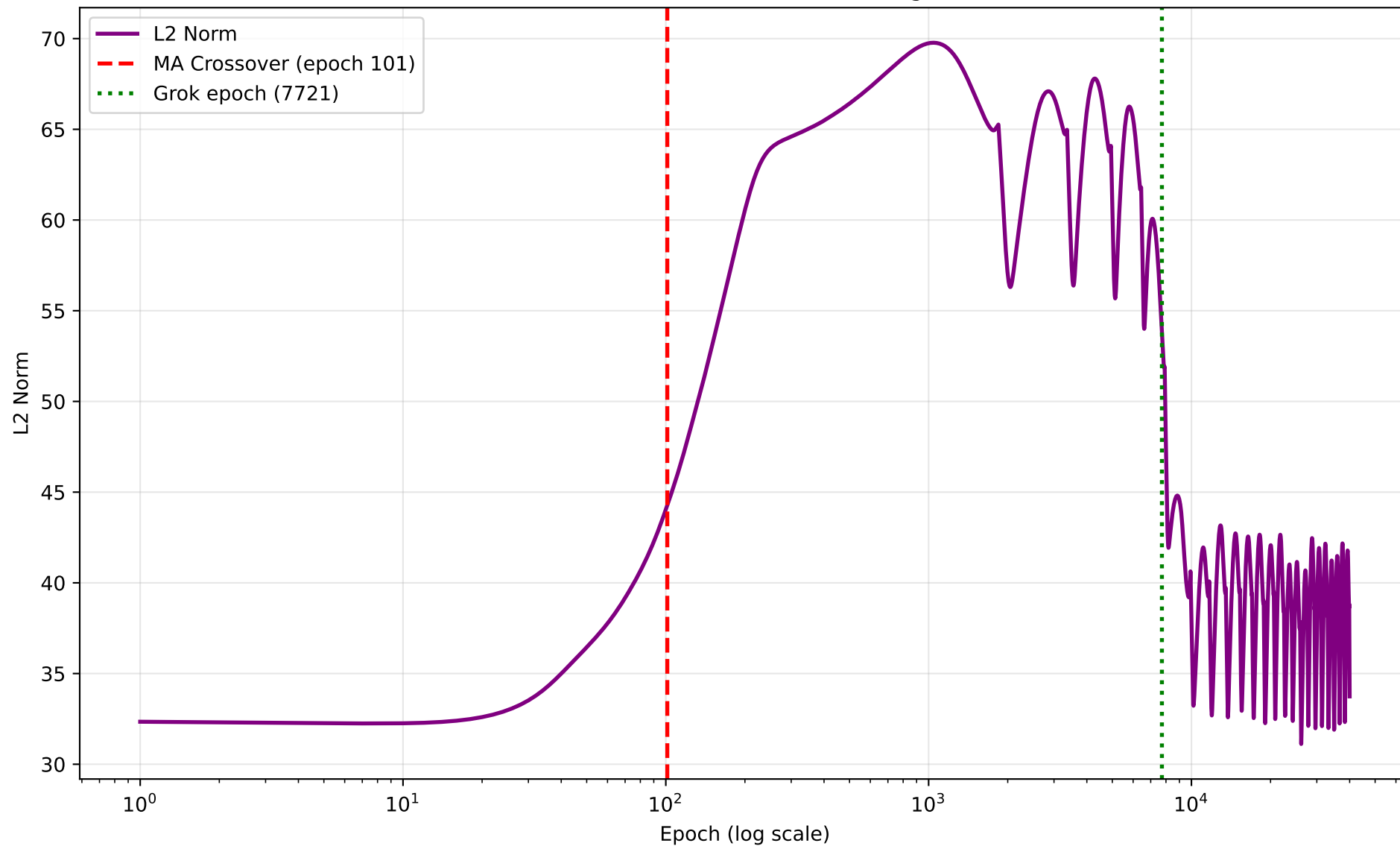
seed 1 — Grokking Curve (grok epoch 7721)



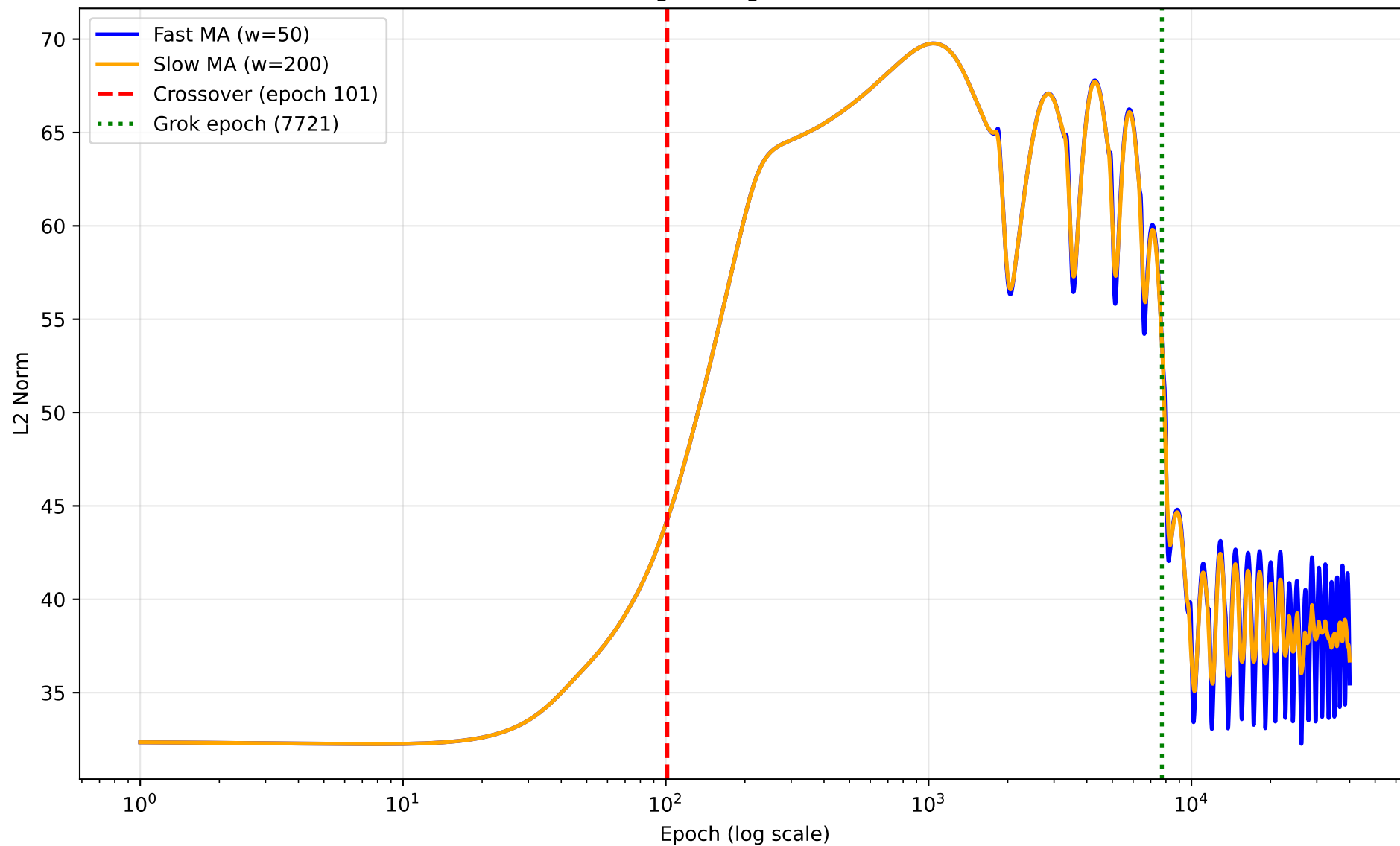
seed 1 — Training Loss



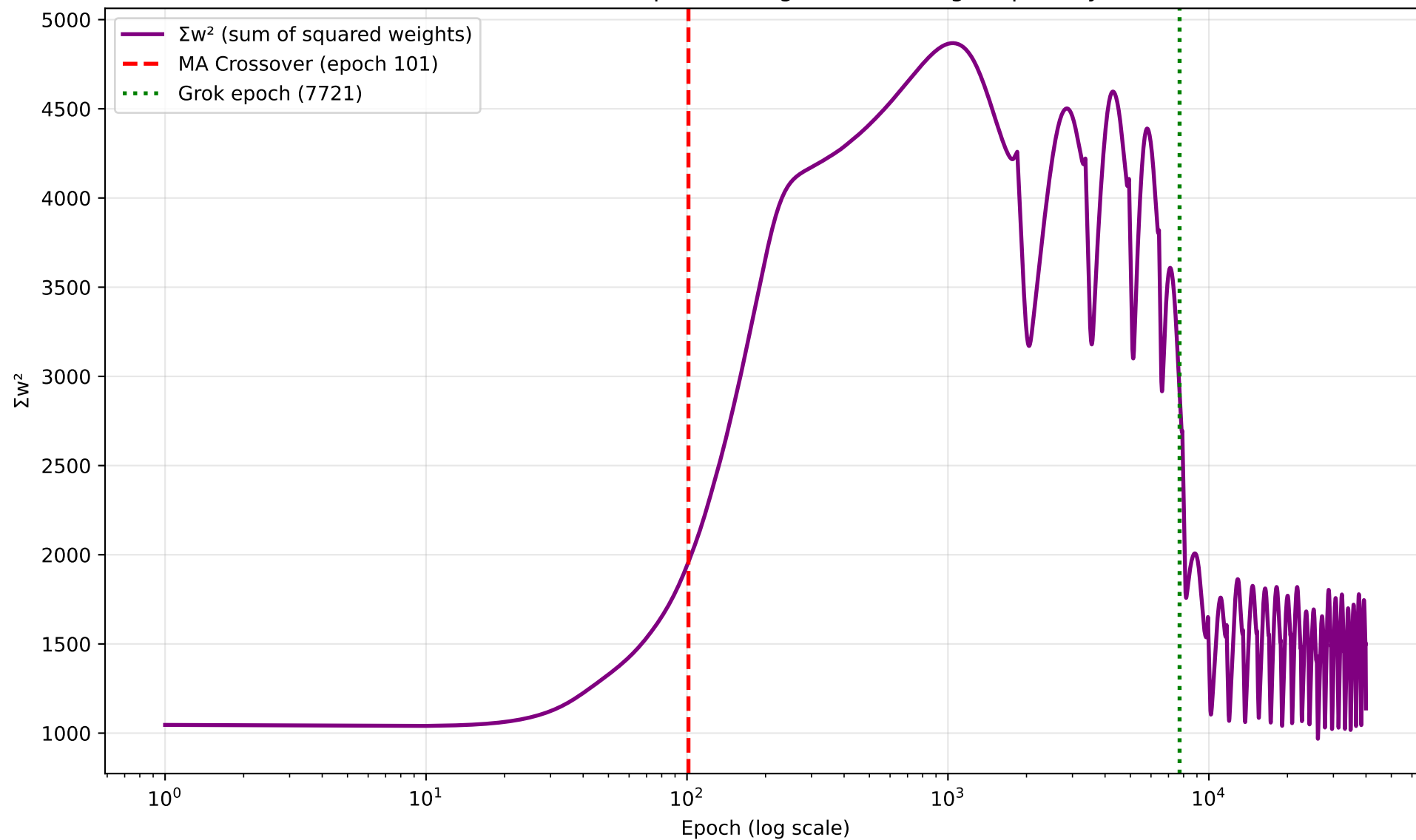
seed 1 — L2 Norm of Model Weights



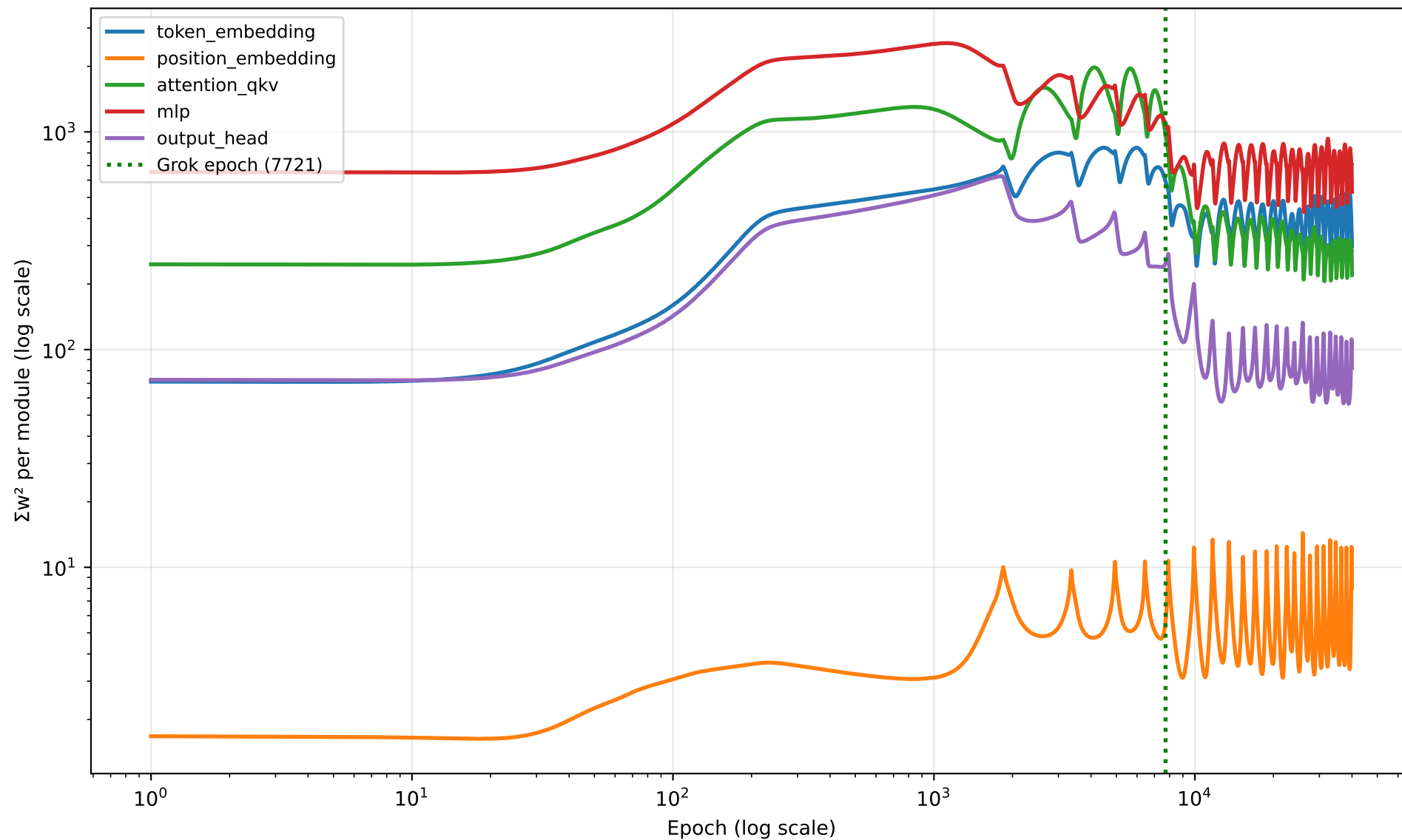
seed 1 — Moving Average Crossover Detection



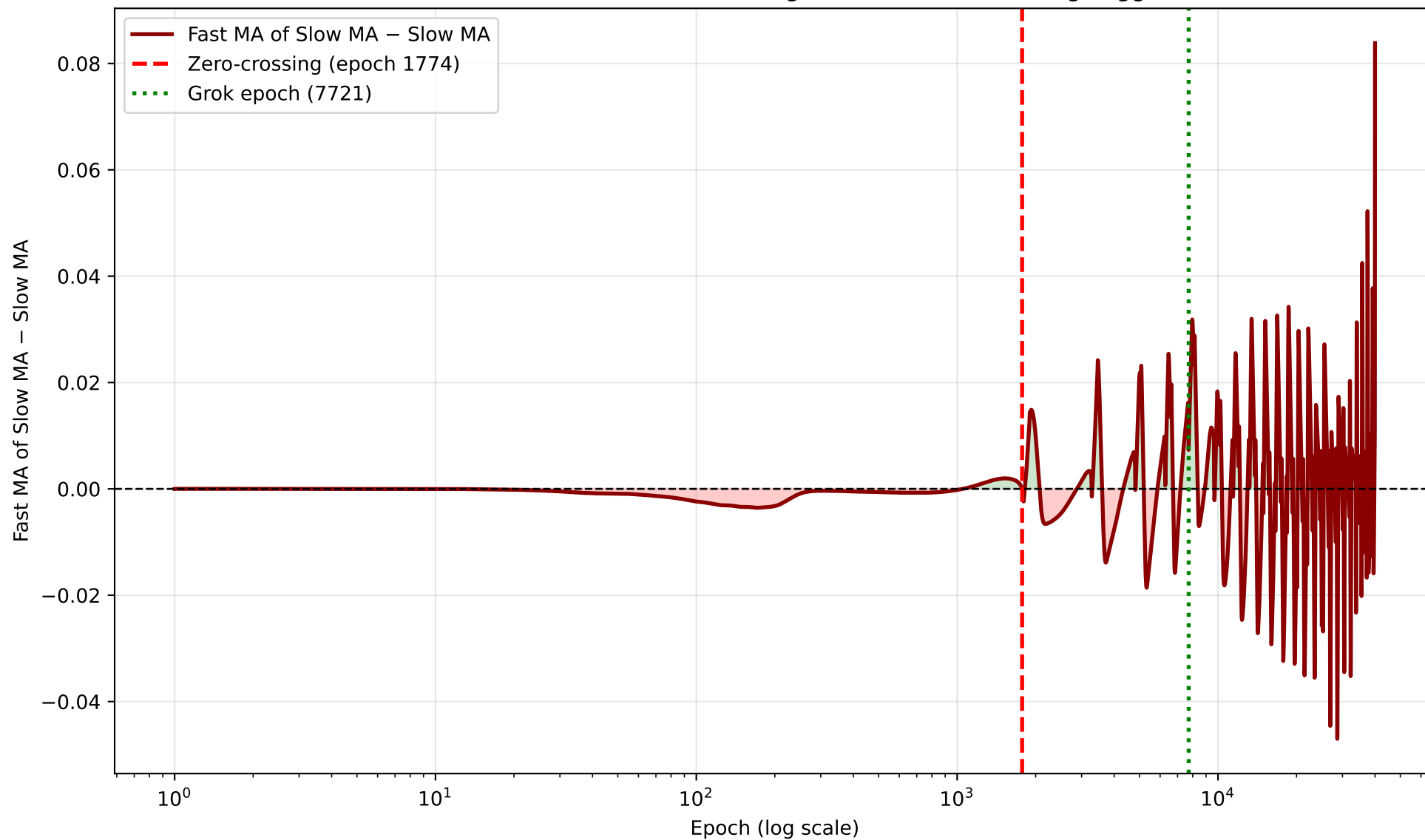
seed 1 — Sum of Squared Weights (Nanda Fig. 7 quantity)



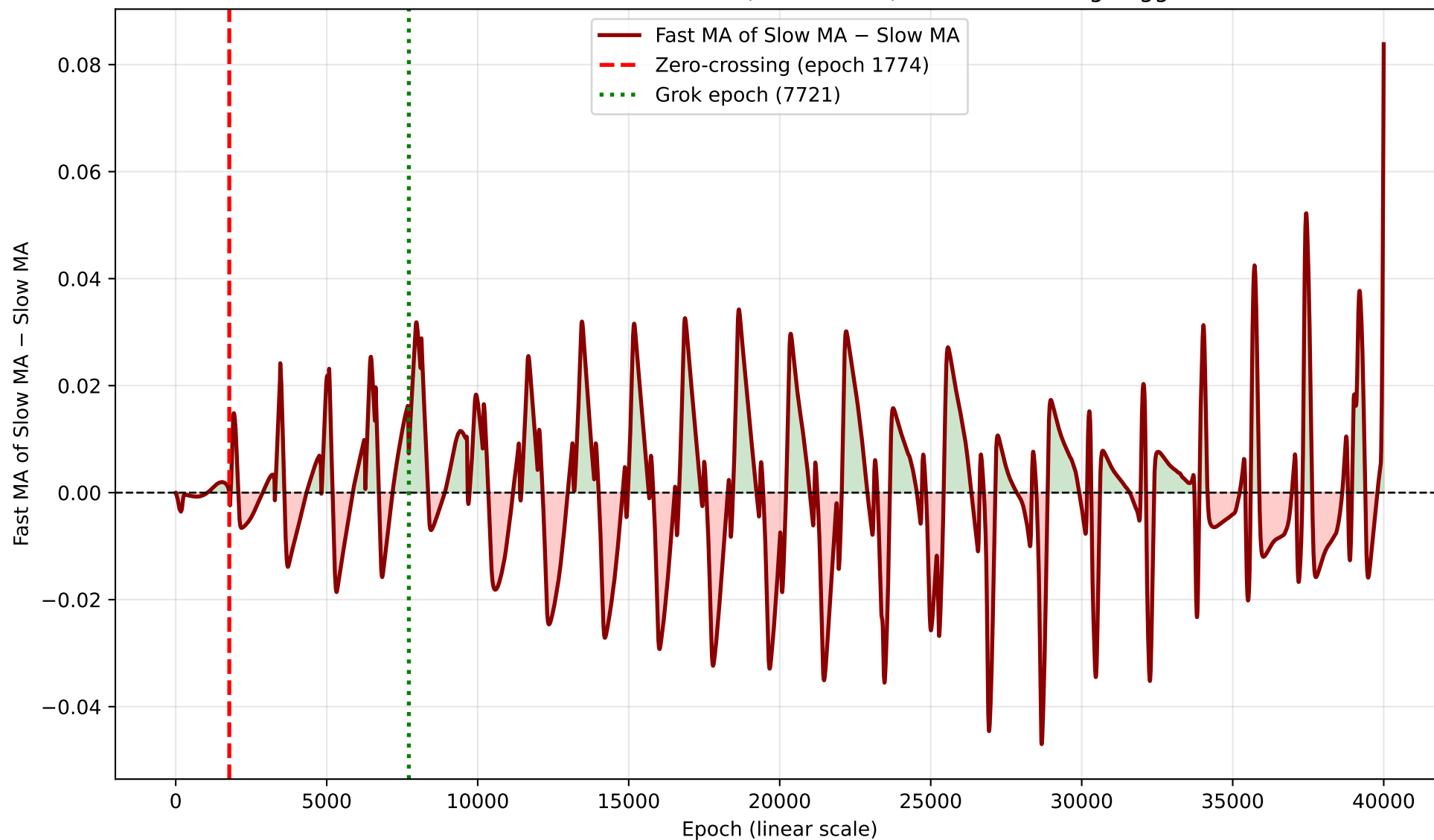
seed 1 — Per-Module Σw^2 Breakdown

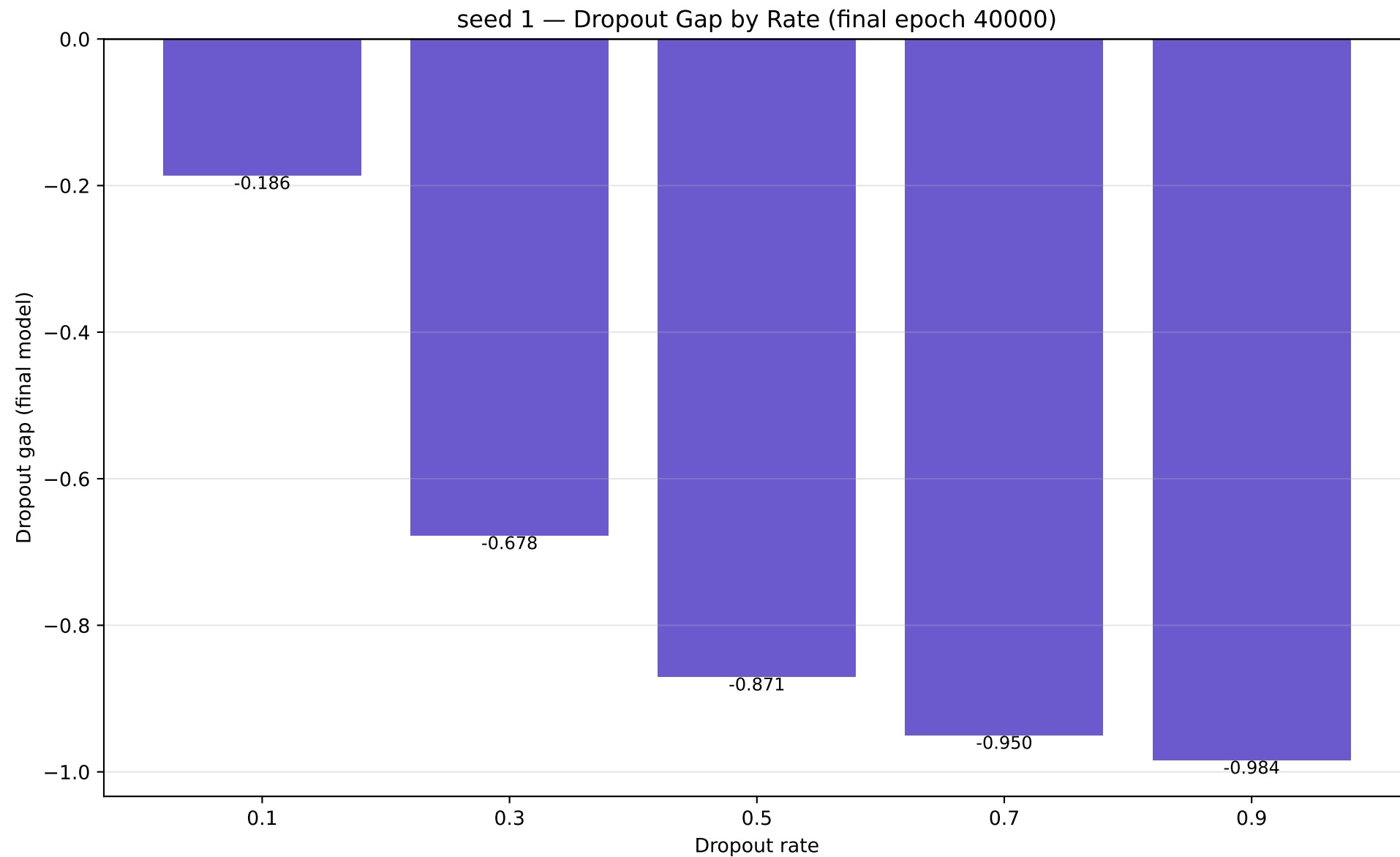


seed 1 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger



seed 1 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger





seed 2 – full numbers

seed 2 (modulus=113 epochs=40000 wall_time=4910.3s)

```
grok_epoch           : 11376
final_train_acc      : 1.0000
final_test_acc       : 1.0000
l2_norm   init -> final : 32.3136 -> 34.7780
sum_w2   init -> final  : 1044.168 -> 1209.512
token_embedding_share (init) : 0.0693
```

L2 predictor:

```
MA-crossover epoch      : 107.21
MA-of-MA zero-crossing epoch : 1833.53
noise_floor             : 0.000465
windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90
```

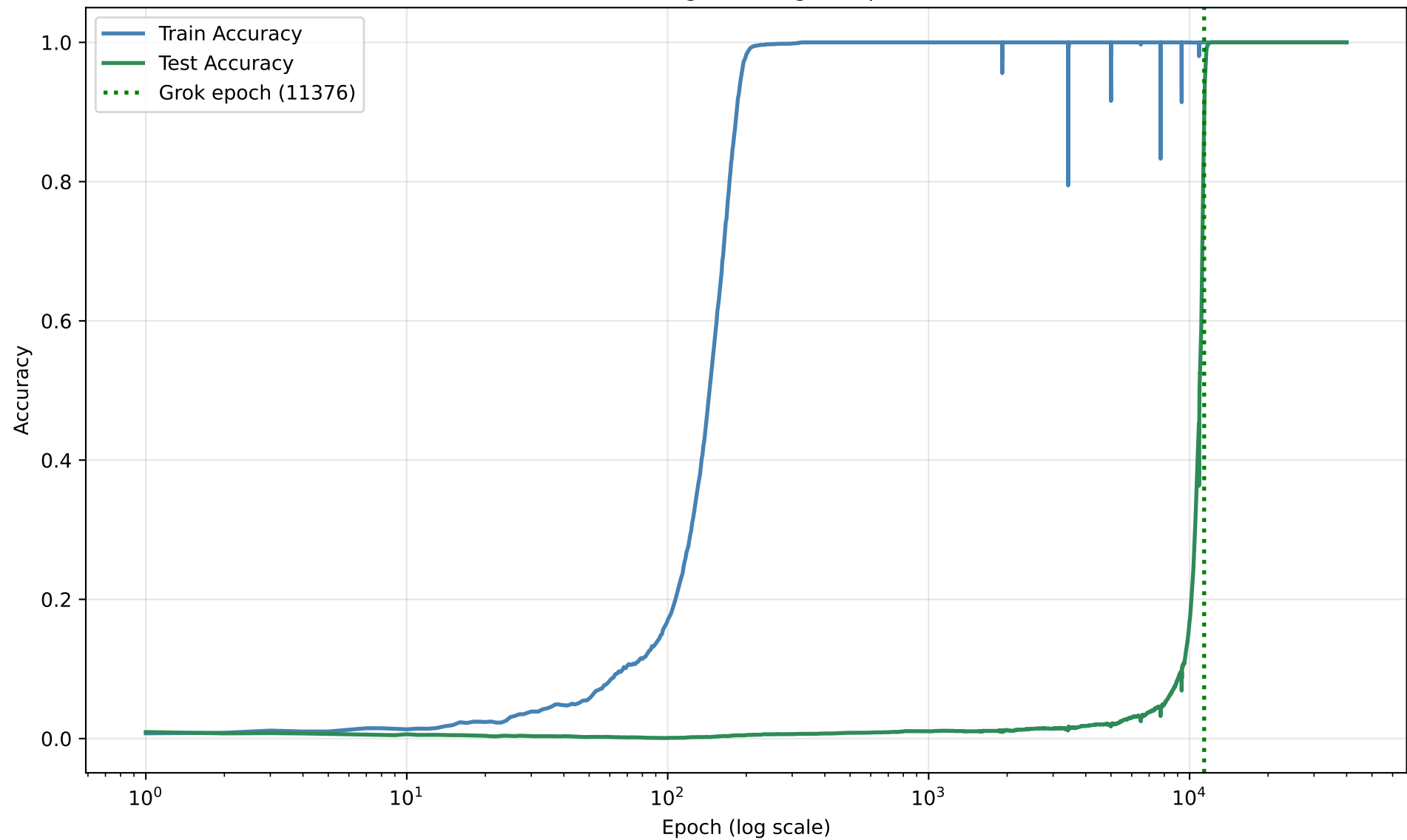
Dropout gap by rate:

```
p0.1=+0.0000 p0.3=-0.0261 p0.5=-0.2475 p0.7=-0.6675 p0.9=-0.9542
```

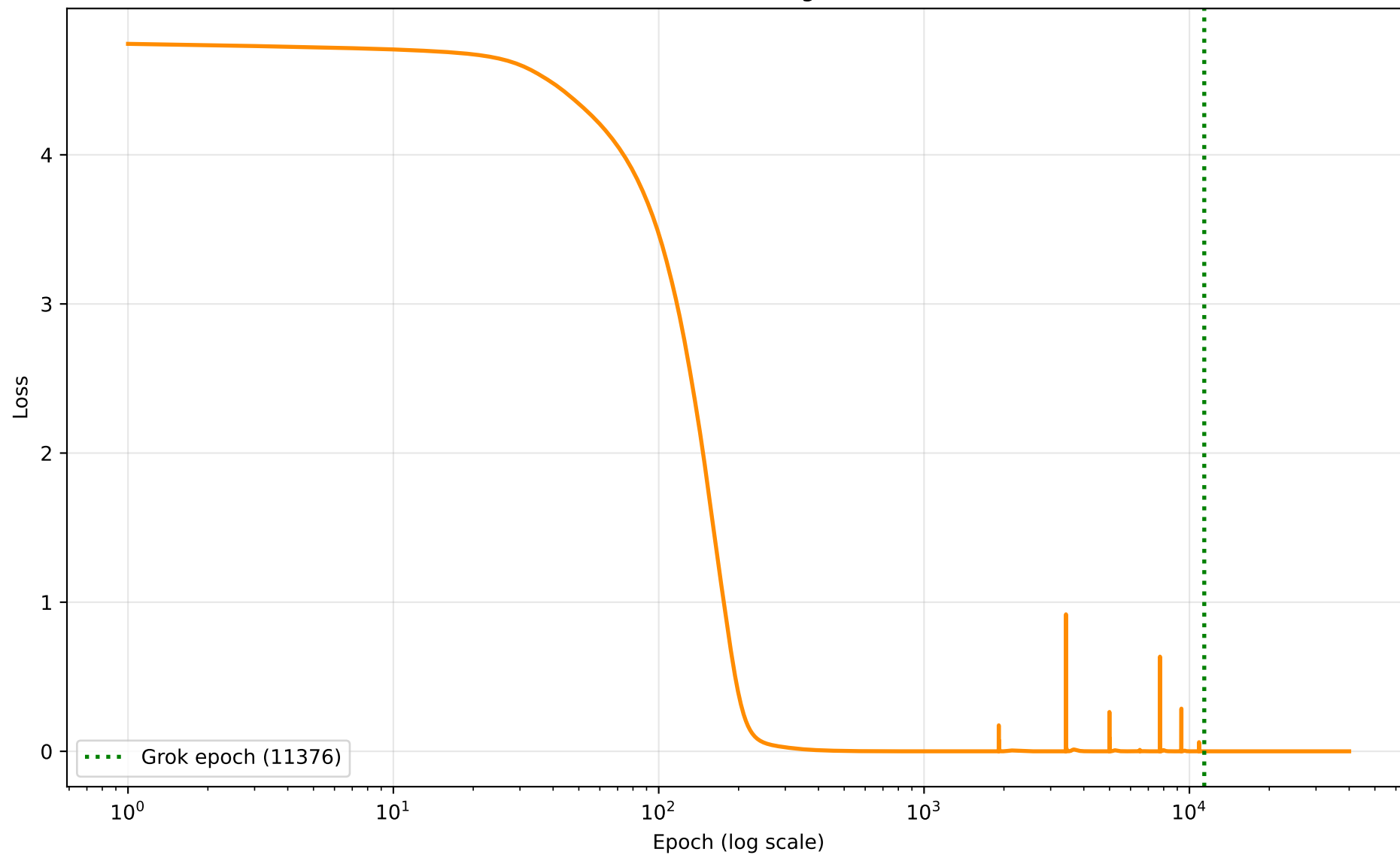
Limit-cycle check : stable

```
window_start=11876 post_grok_min=0.9990 post_grok_std=0.0001
post_grok_final=1.0000 dips<0.9=0
```

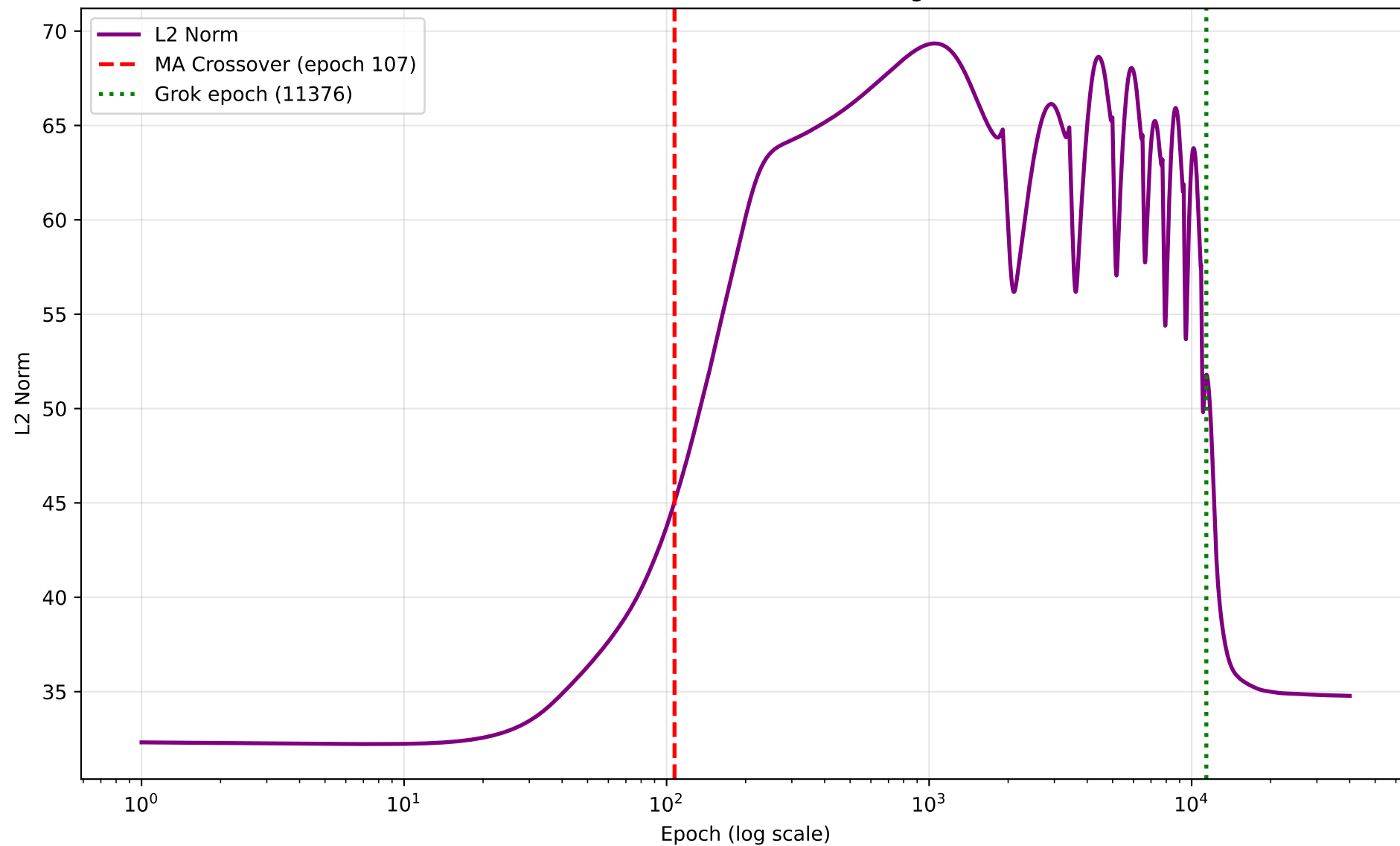
seed 2 — Grokking Curve (grok epoch 11376)



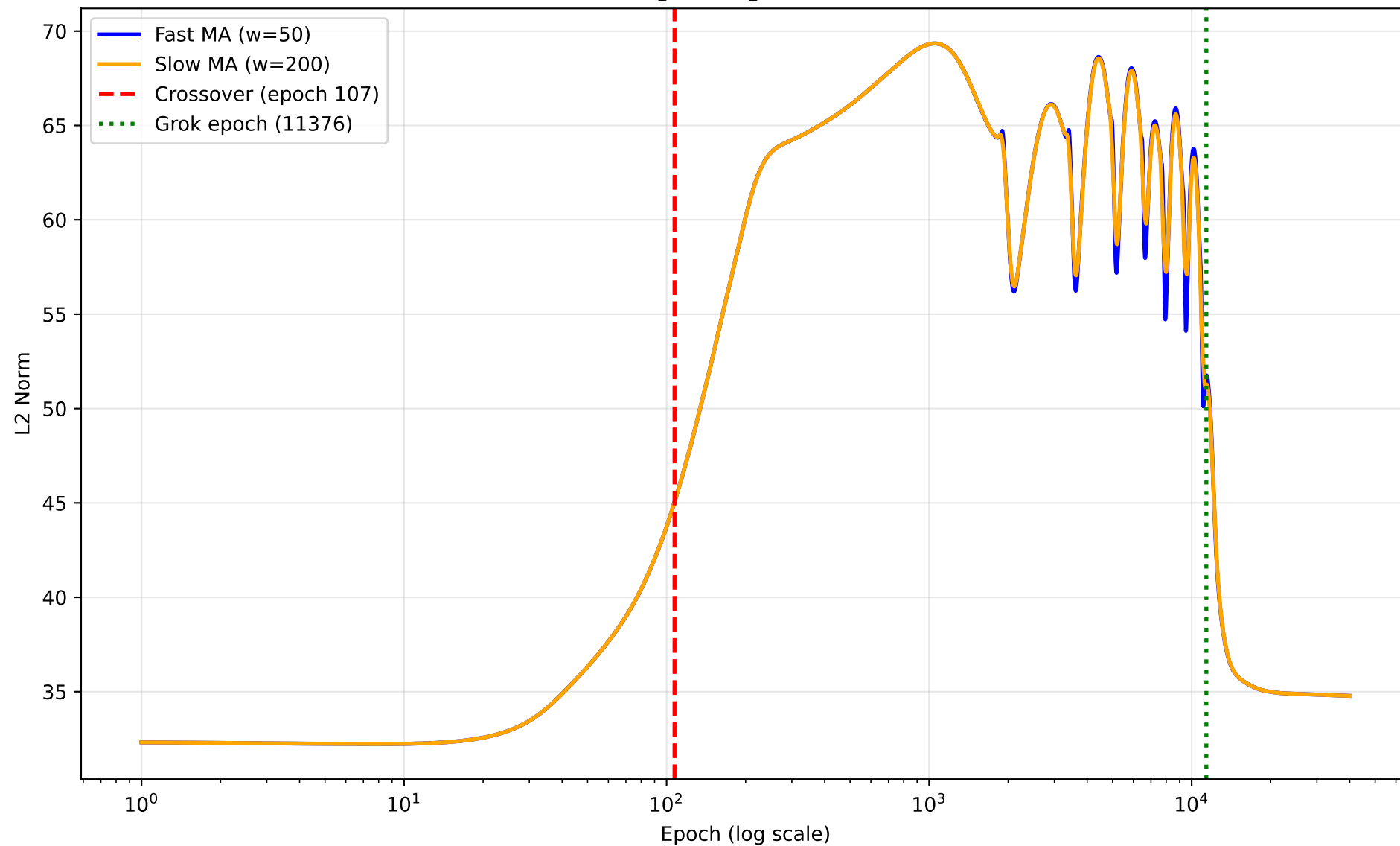
seed 2 — Training Loss



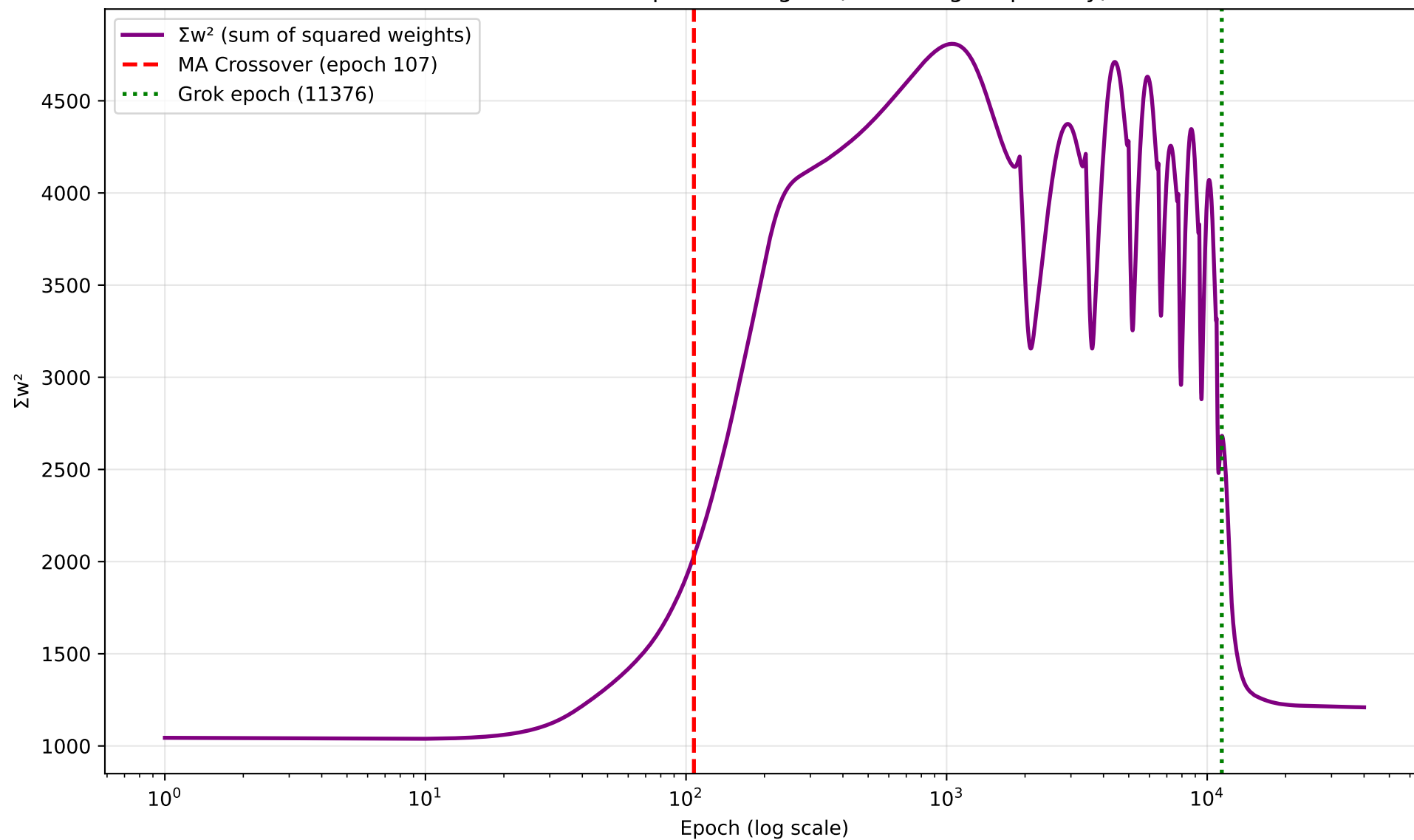
seed 2 — L2 Norm of Model Weights



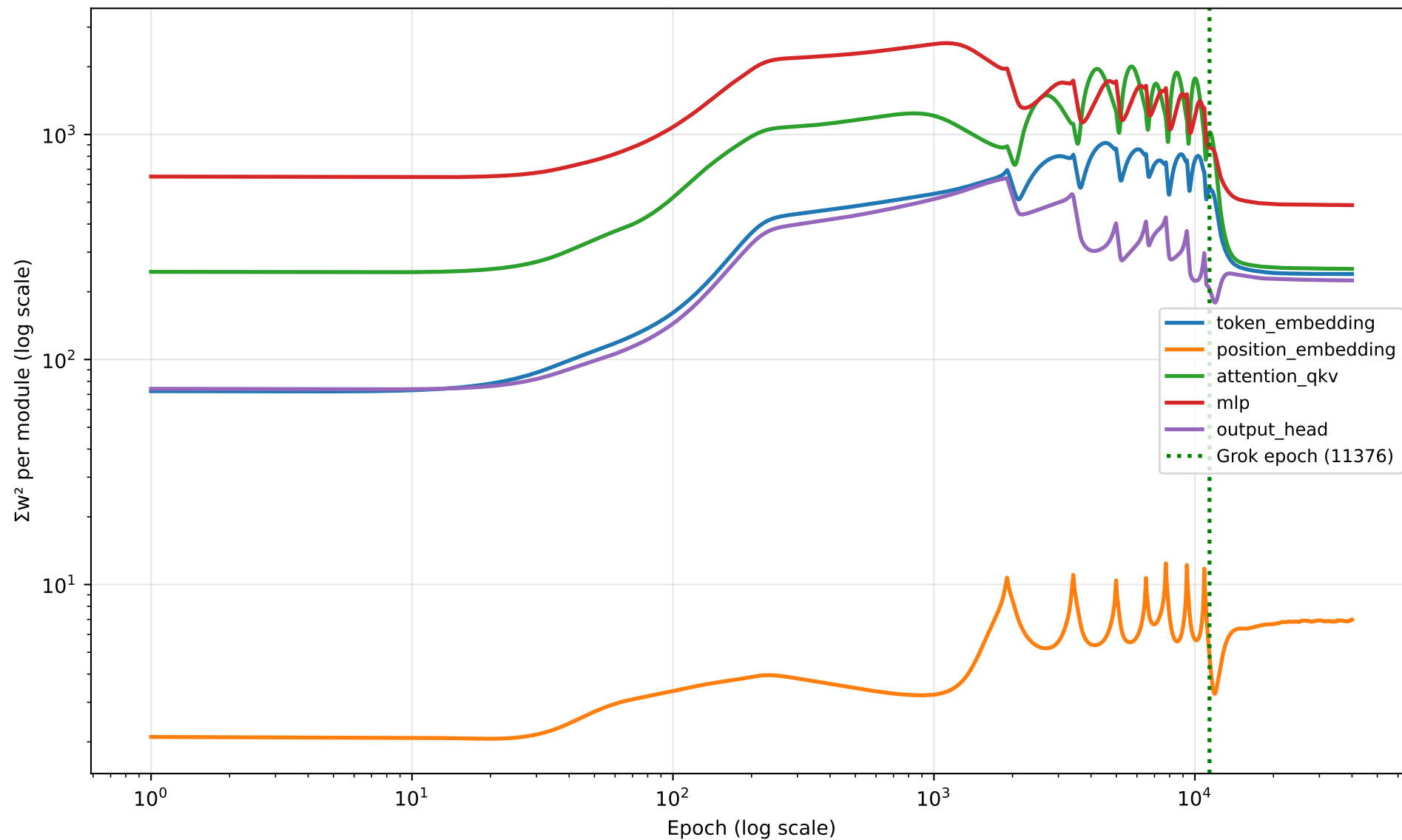
seed 2 — Moving Average Crossover Detection



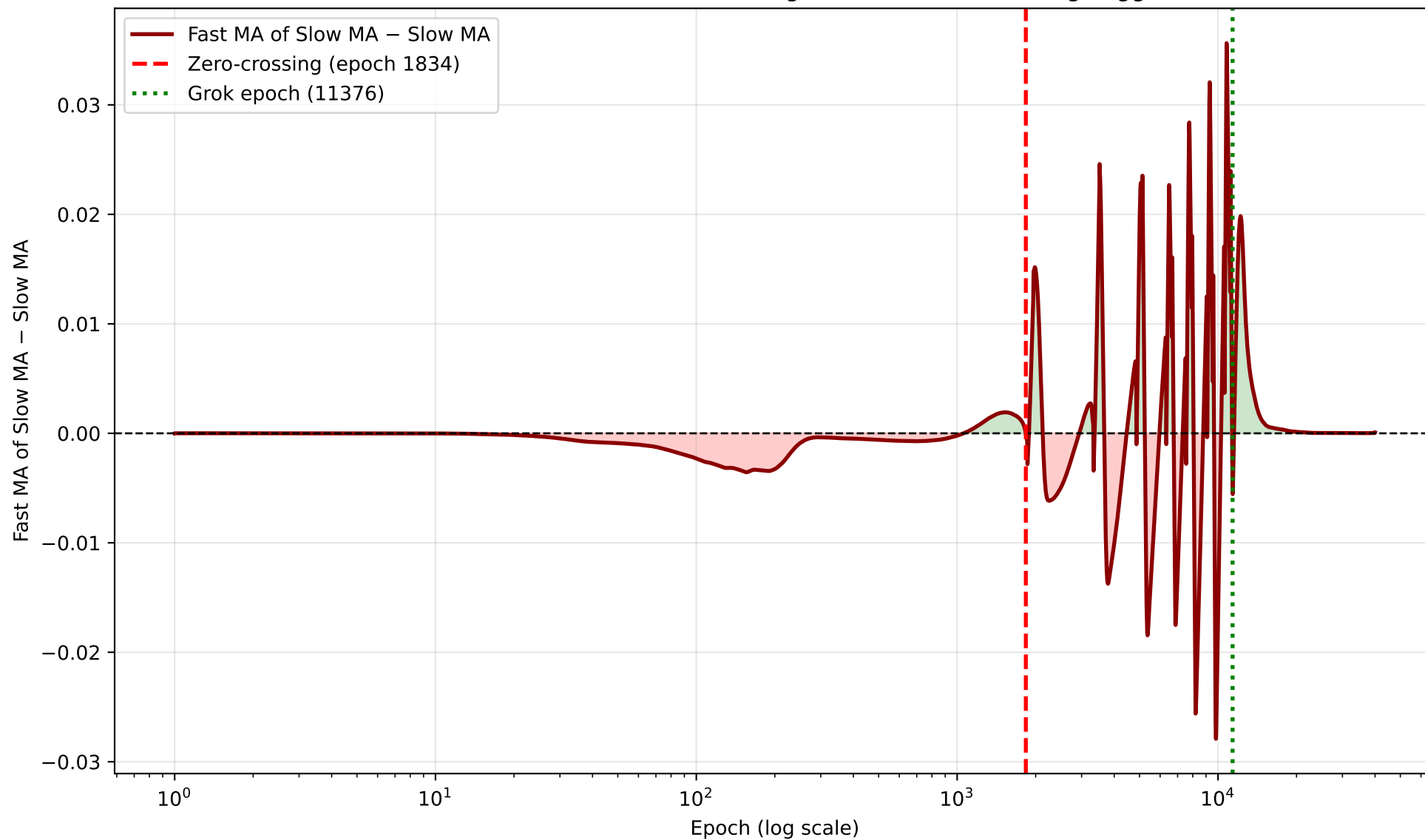
seed 2 — Sum of Squared Weights (Nanda Fig. 7 quantity)



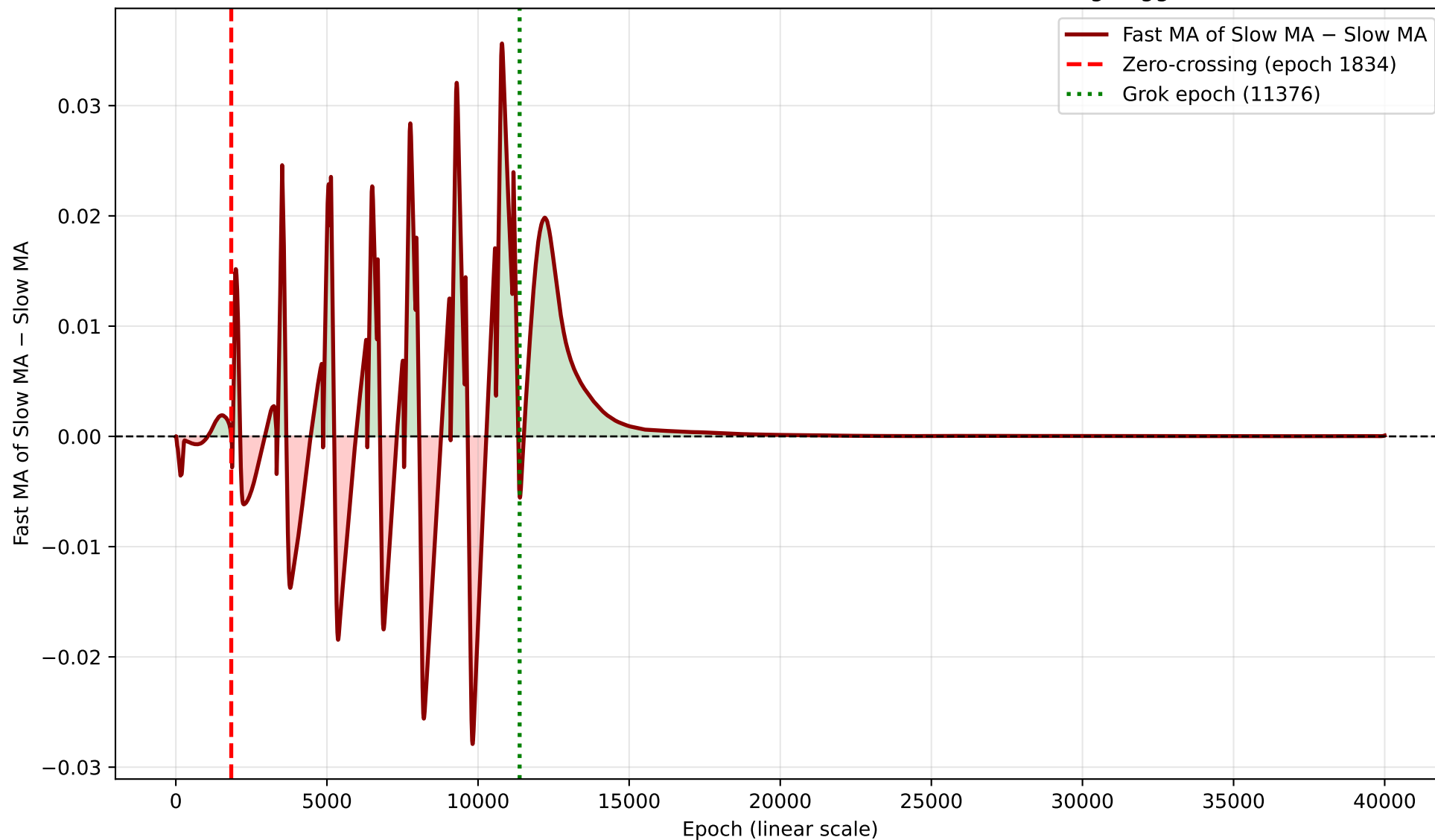
seed 2 — Per-Module Σw^2 Breakdown

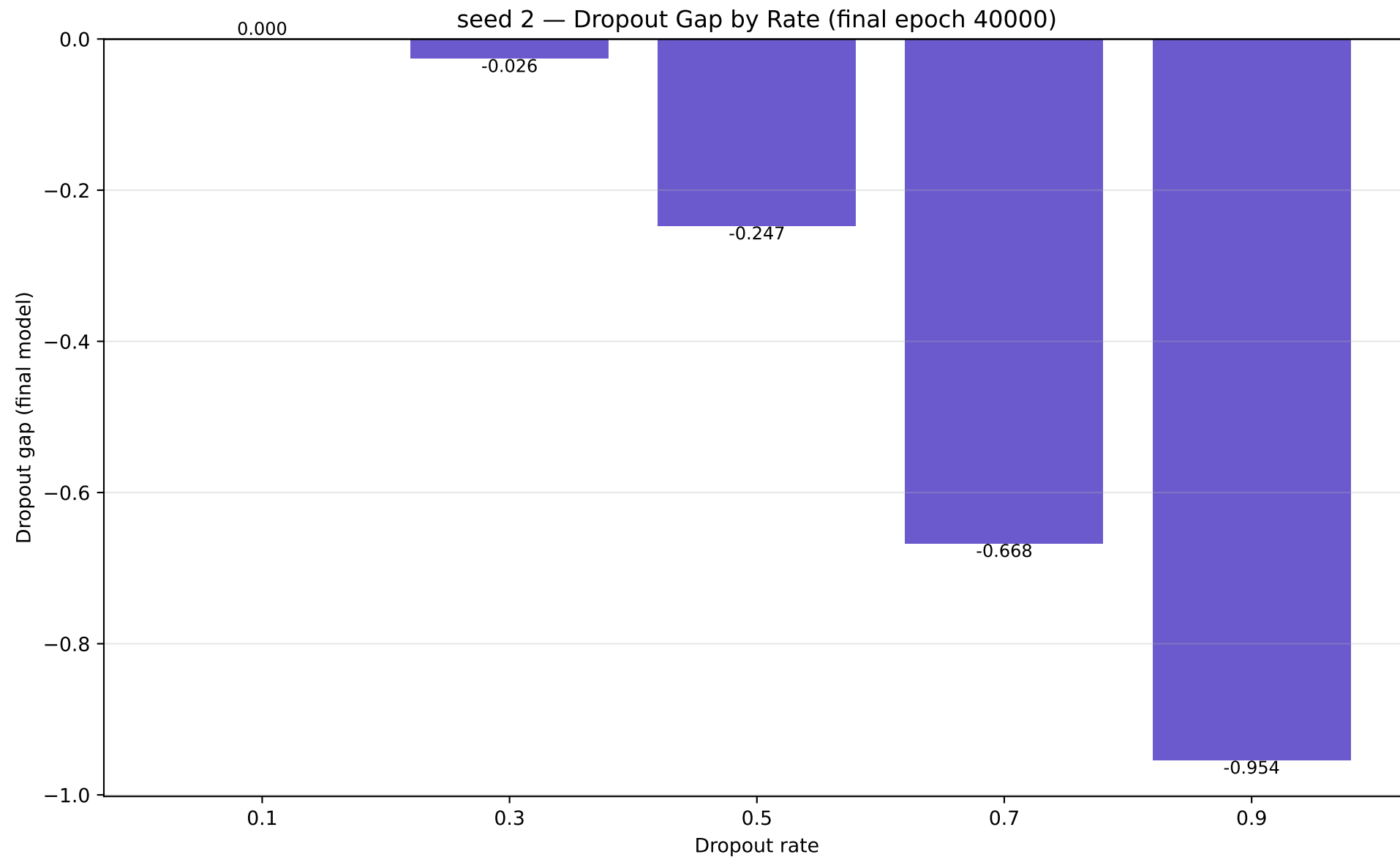


seed 2 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger



seed 2 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger





seed 3 – full numbers

seed 3 (modulus=113 epochs=40000 wall_time=4903.2s)

grok_epoch : 12678
final_train_acc : 1.0000
final_test_acc : 1.0000
l2_norm init -> final : 32.3835 -> 35.9785
sum_w2 init -> final : 1048.690 -> 1294.450
token_embedding_share (init) : 0.0680

L2 predictor:

MA-crossover epoch : 114.01
MA-of-MA zero-crossing epoch : 1798.41
noise_floor : 0.000462
windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

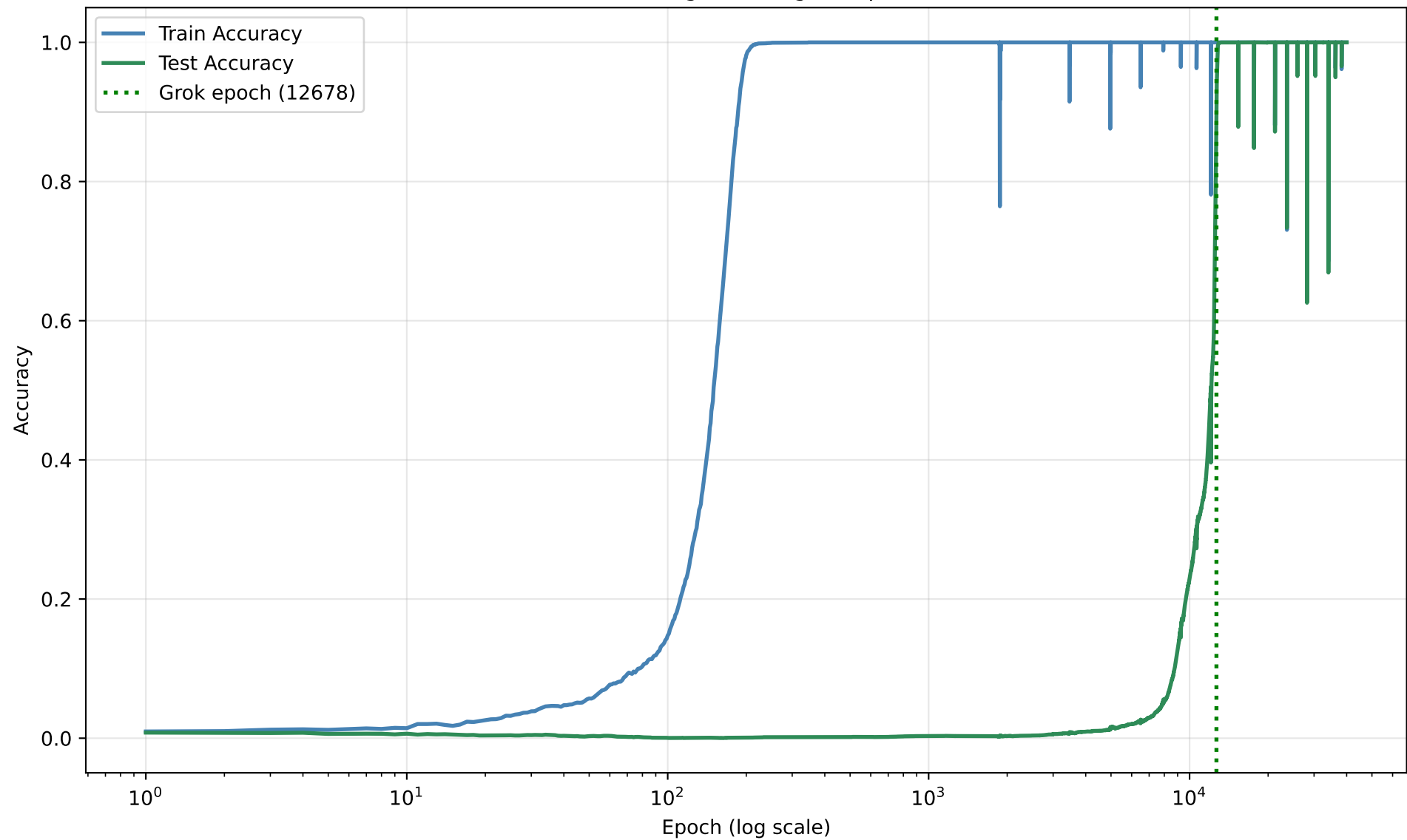
Dropout gap by rate:

p0.1=-0.0009 p0.3=-0.0847 p0.5=-0.4772 p0.7=-0.8438 p0.9=-0.9791

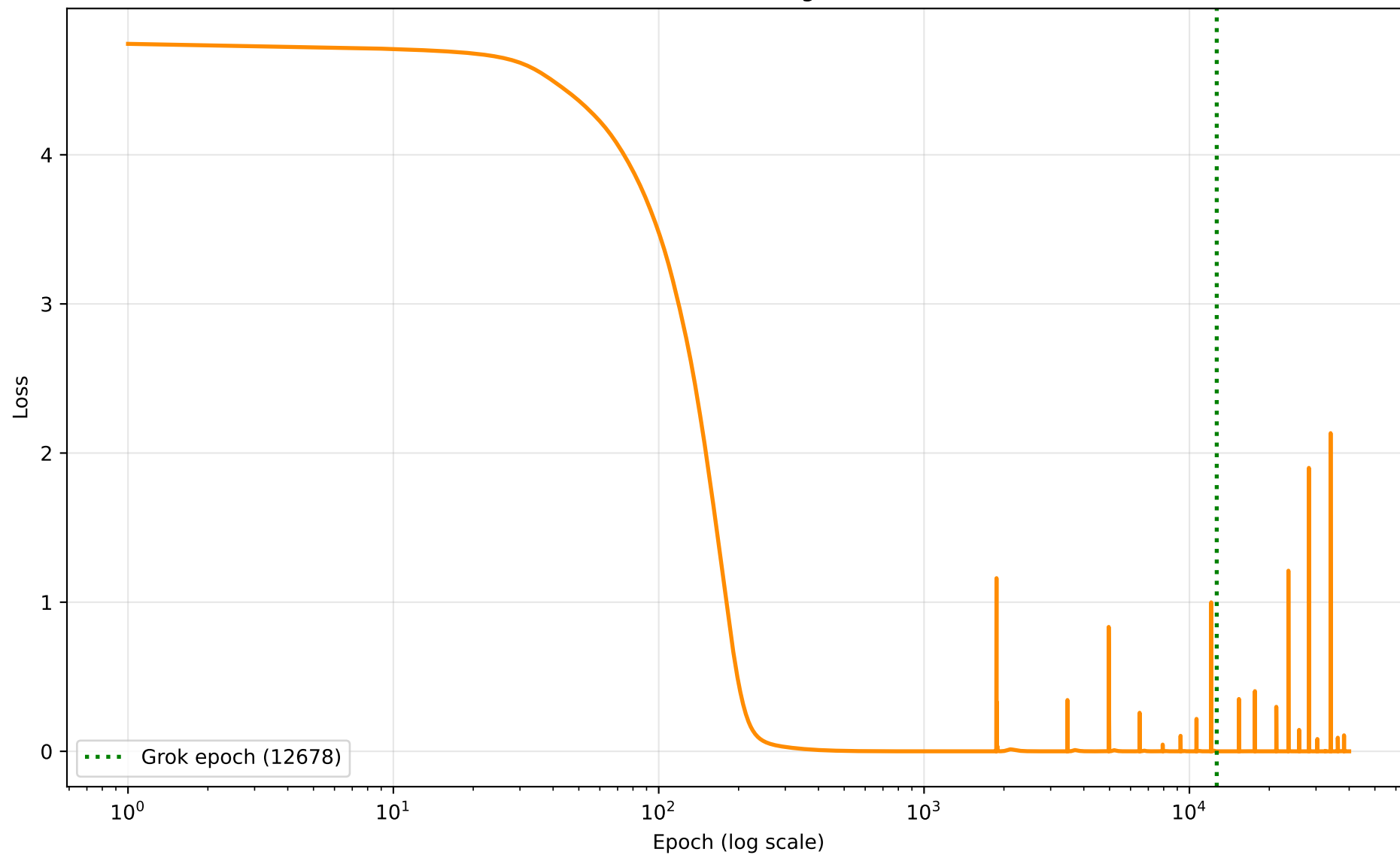
Limit-cycle check : stable

window_start=13178 post_grok_min=0.6258 post_grok_std=0.0056
post_grok_final=1.0000 dips<0.9=17

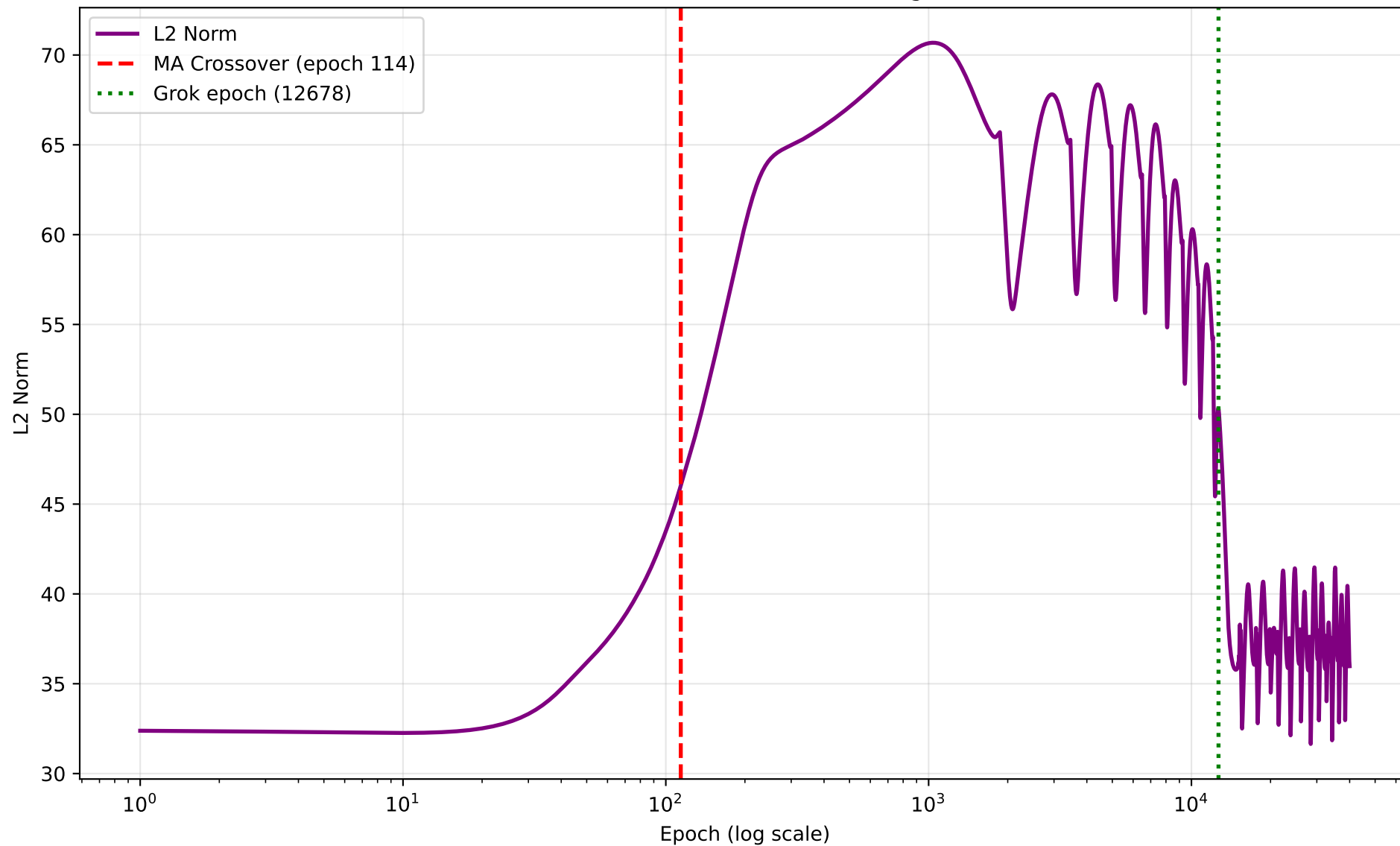
seed 3 — Grokking Curve (grok epoch 12678)



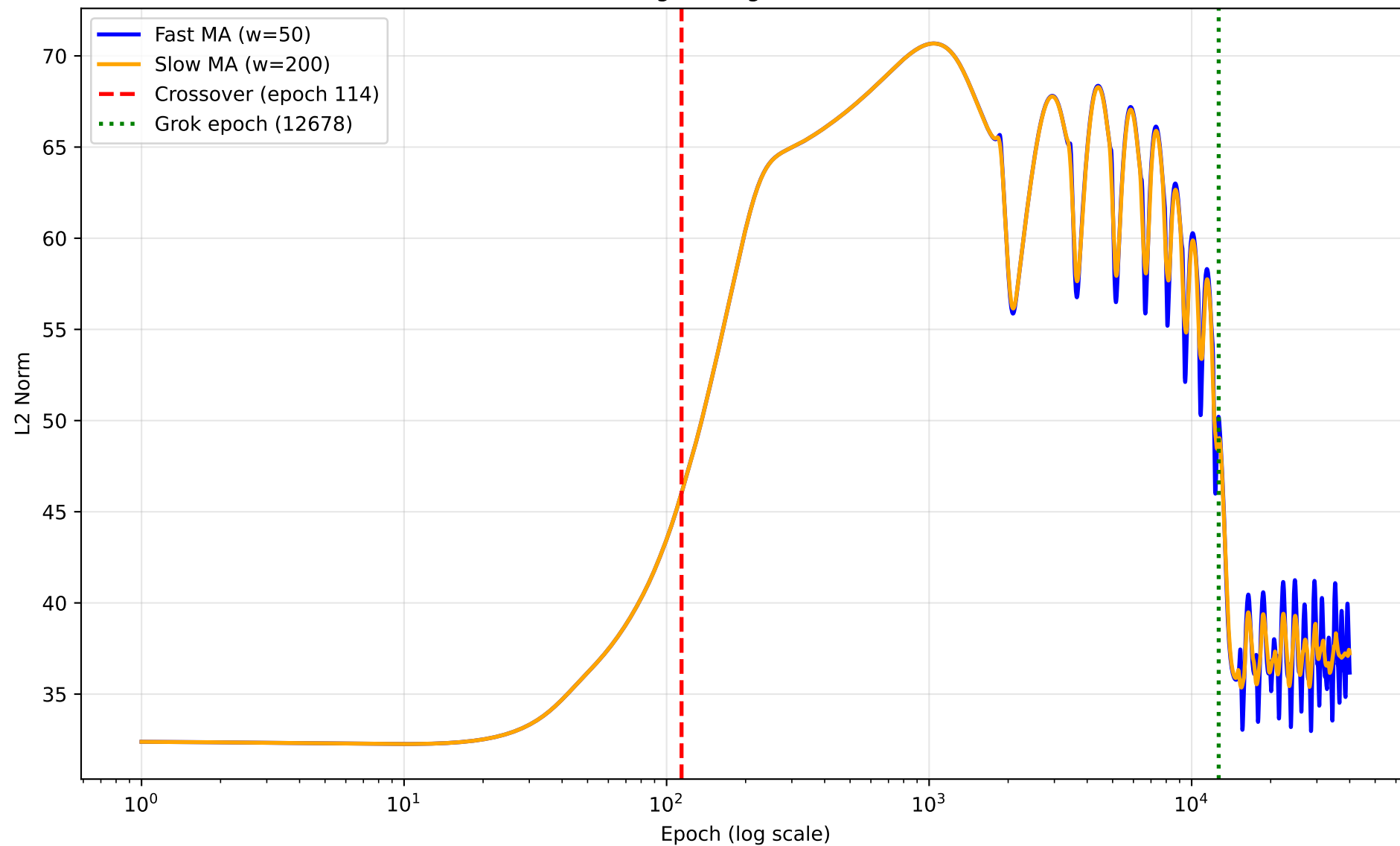
seed 3 — Training Loss



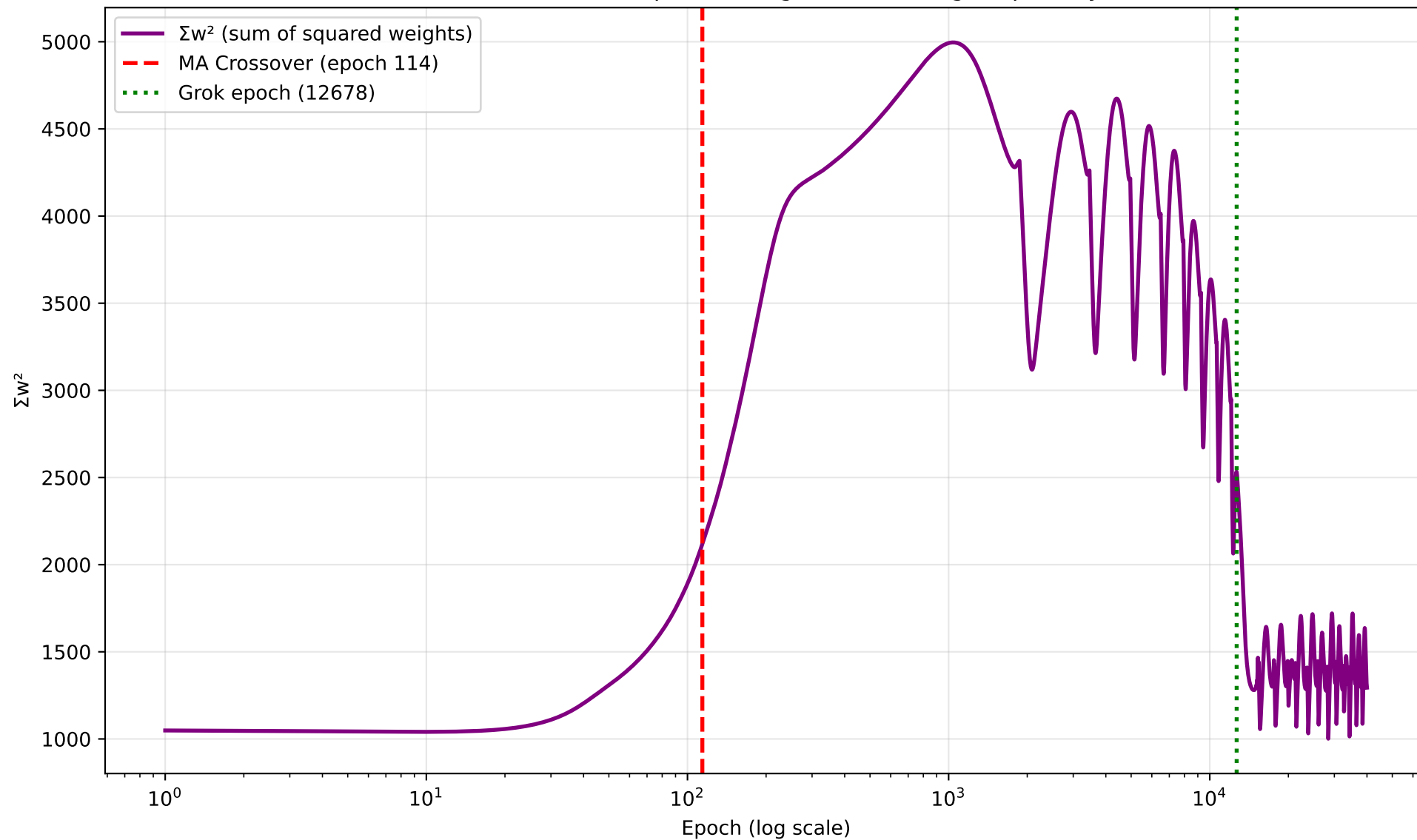
seed 3 — L2 Norm of Model Weights



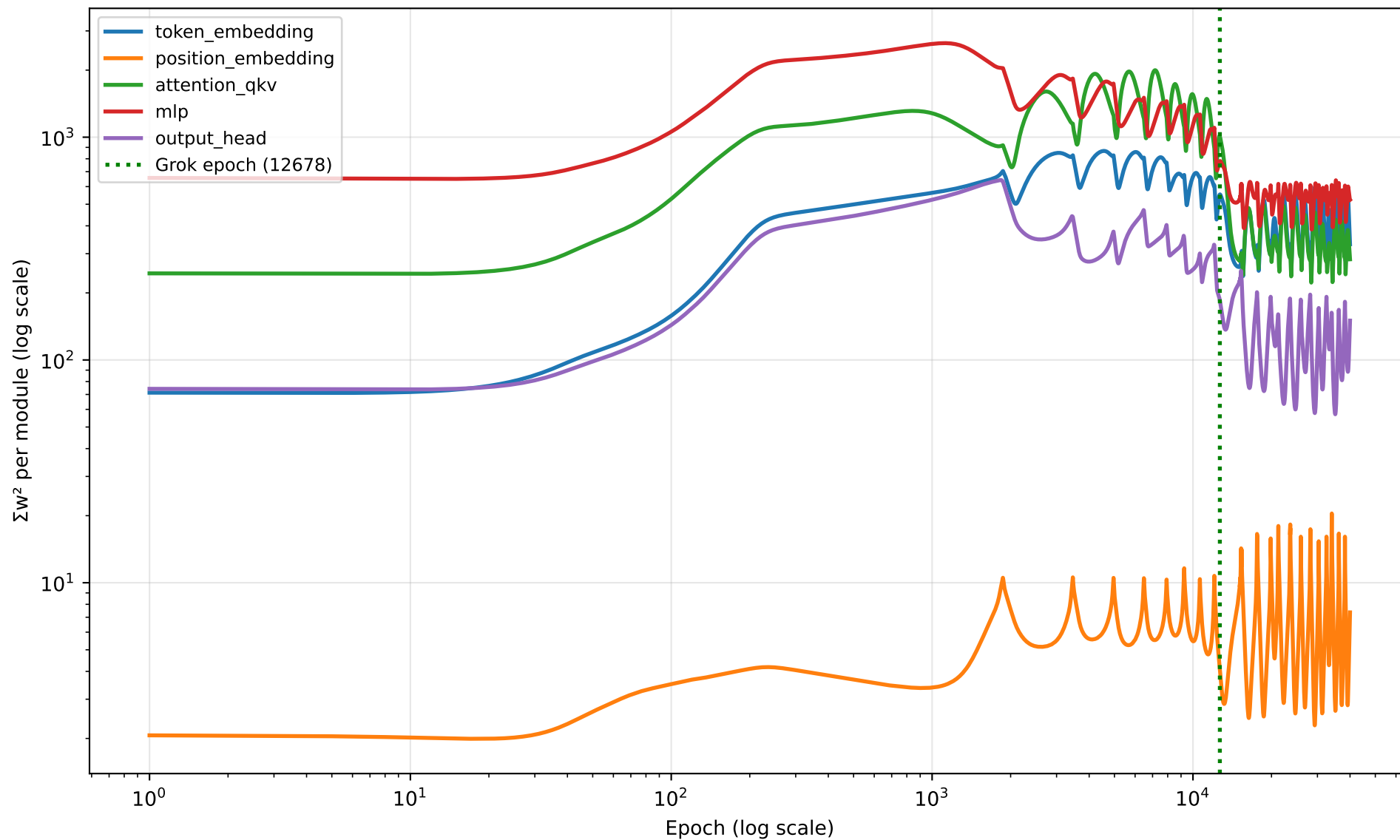
seed 3 — Moving Average Crossover Detection



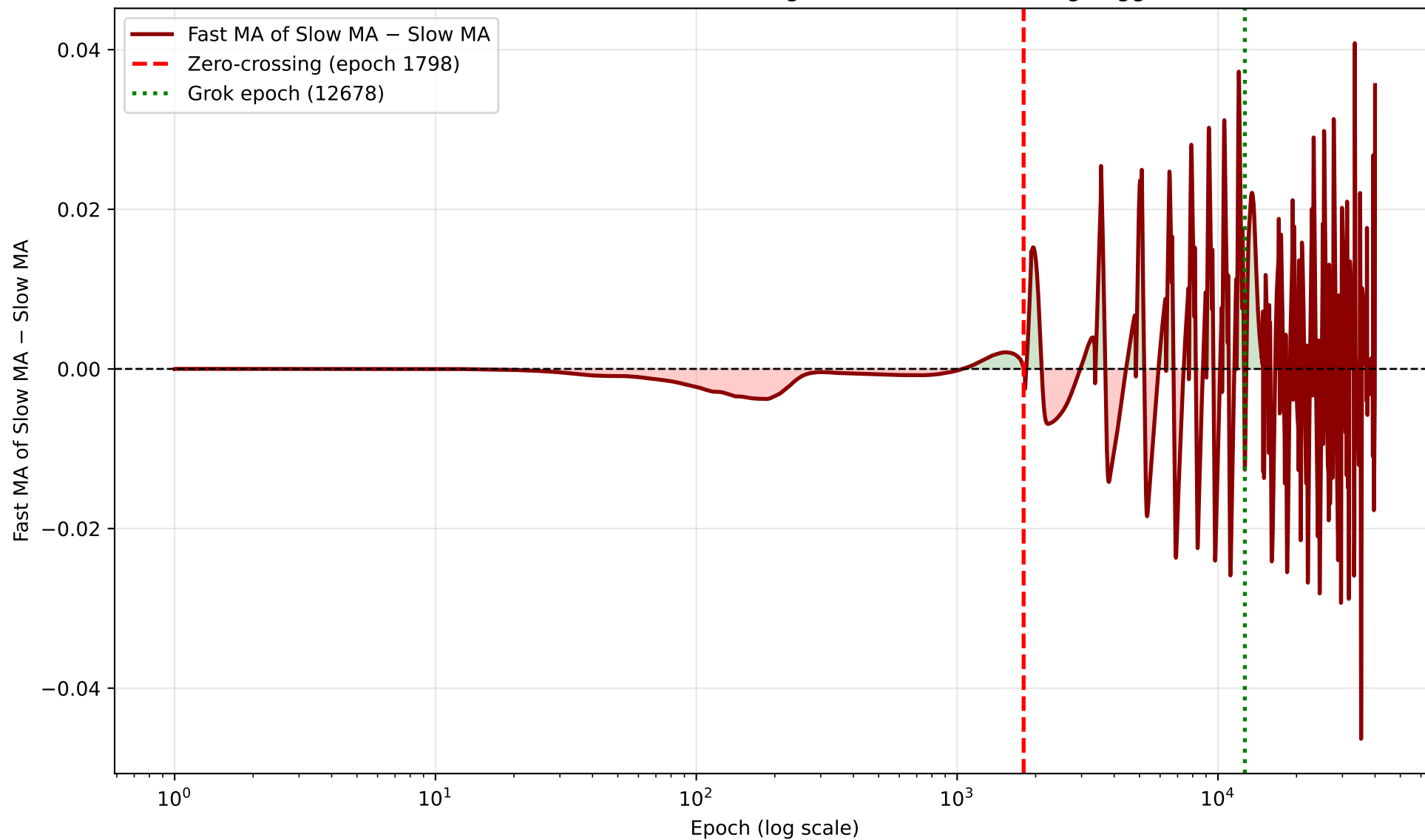
seed 3 — Sum of Squared Weights (Nanda Fig. 7 quantity)



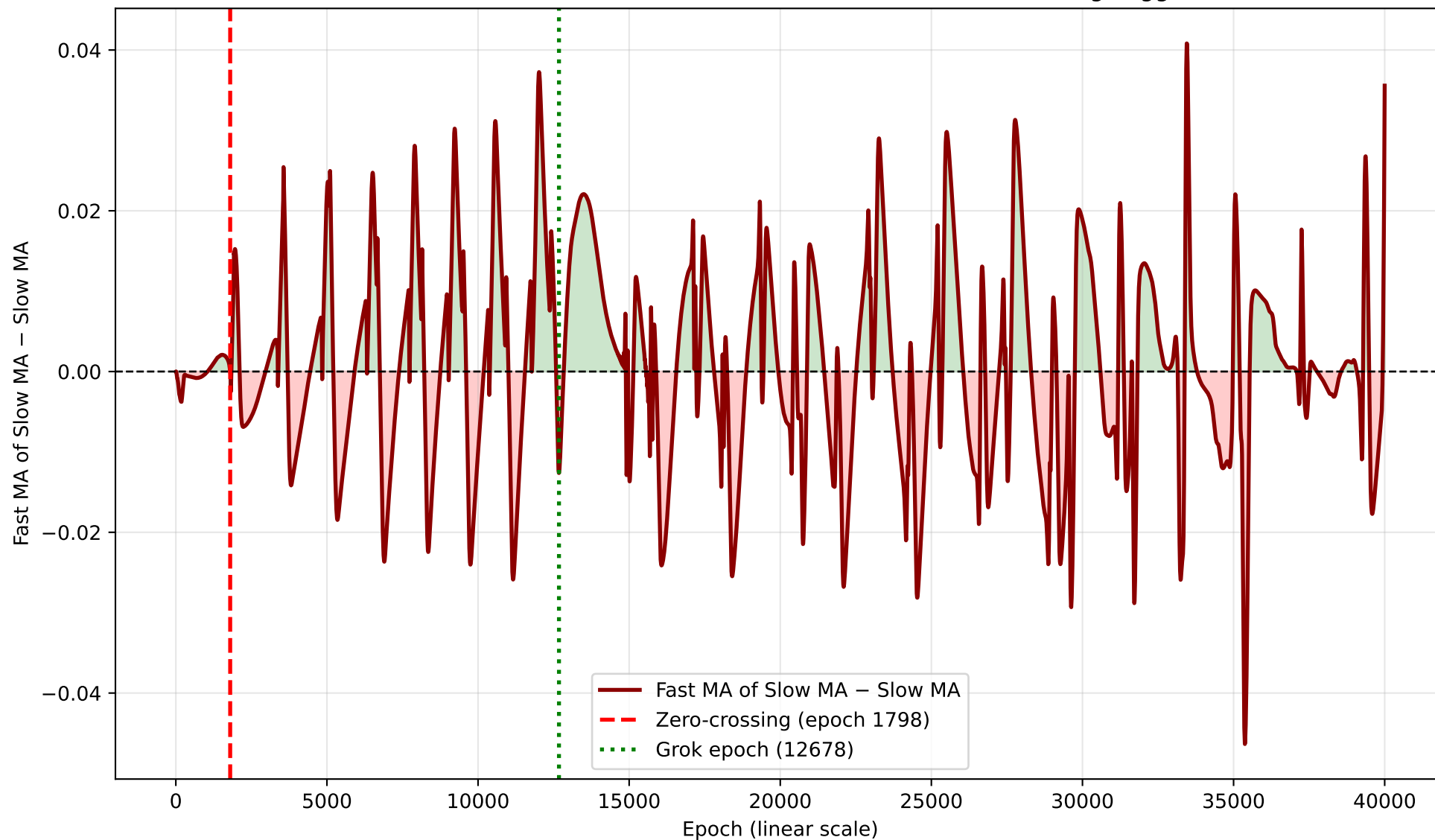
seed 3 — Per-Module Σw^2 Breakdown

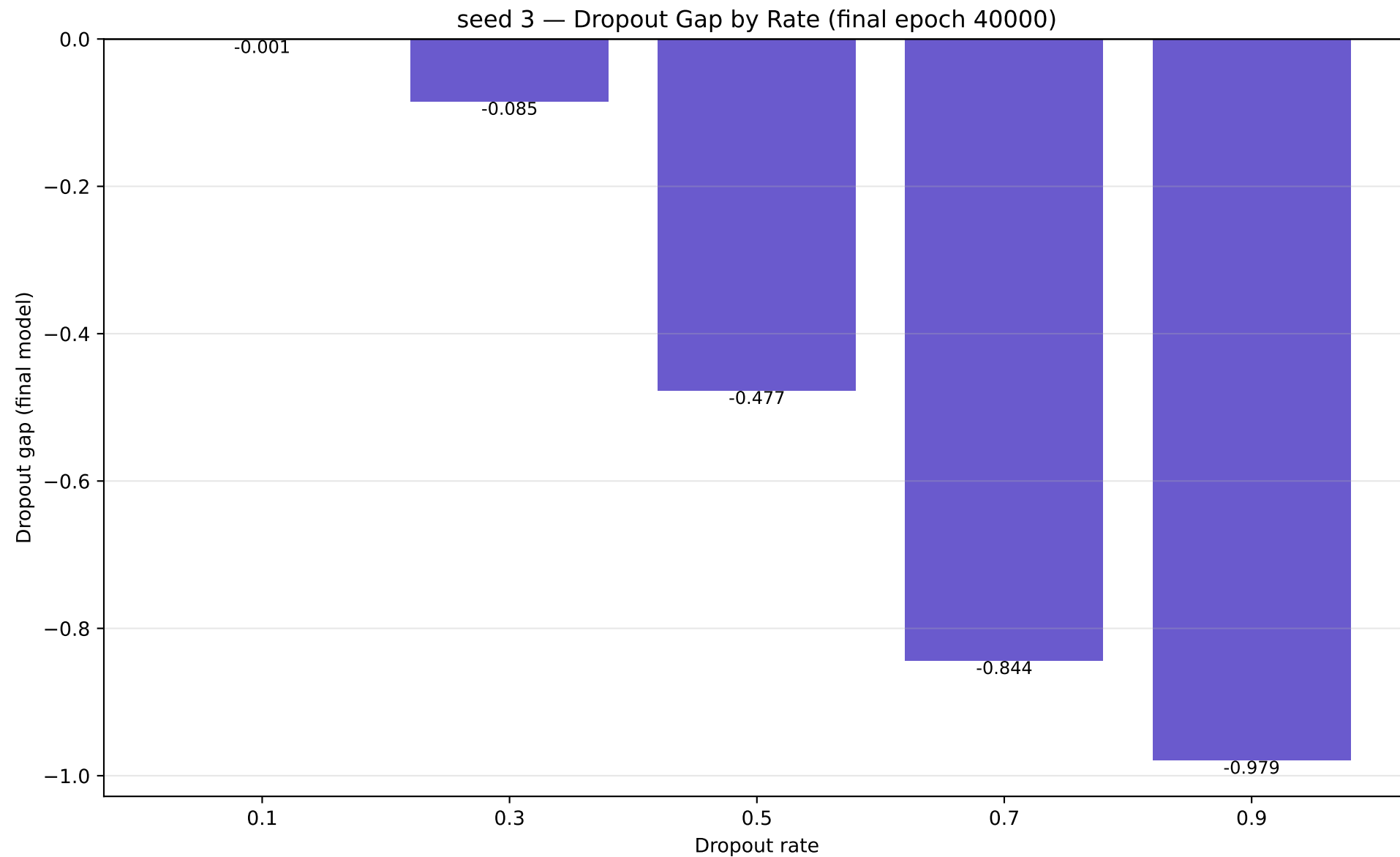


seed 3 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger



seed 3 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger





seed 4 – full numbers

seed 4 (modulus=113 epochs=40000 wall_time=4932.6s)

grok_epoch : 25505
final_train_acc : 1.0000
final_test_acc : 1.0000
l2_norm init -> final : 32.2933 -> 32.6518
sum_w2 init -> final : 1042.858 -> 1066.140
token_embedding_share (init) : 0.0697

L2 predictor:

MA-crossover epoch : 143.98
MA-of-MA zero-crossing epoch : 1778.98
noise_floor : 0.000469
windows: fast=50 slow=200 ma_of_ma_fast=20 skip_epochs=100 quiet_epoch_cutoff=90

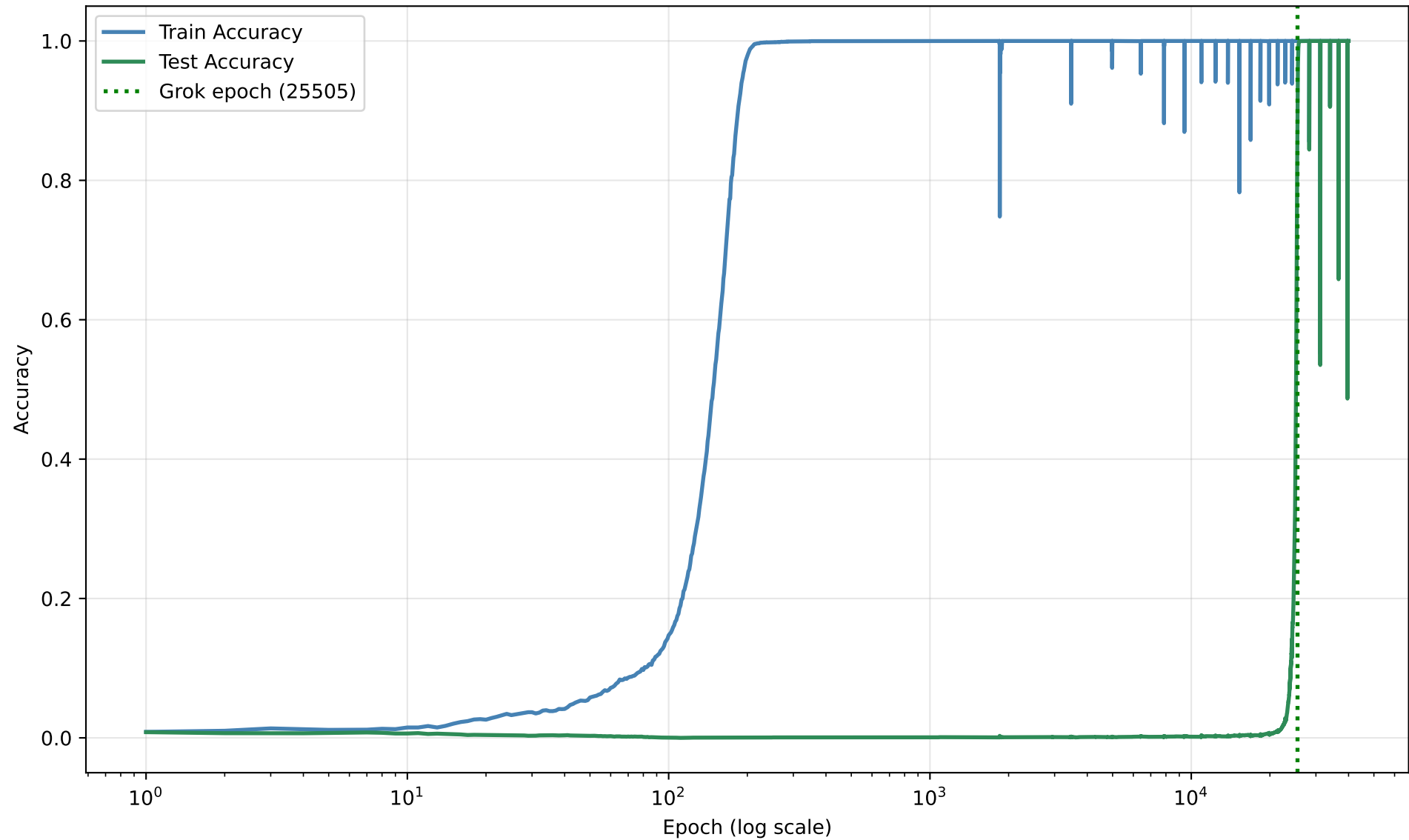
Dropout gap by rate:

p0.1=-0.0126 p0.3=-0.3450 p0.5=-0.7282 p0.7=-0.9246 p0.9=-0.9843

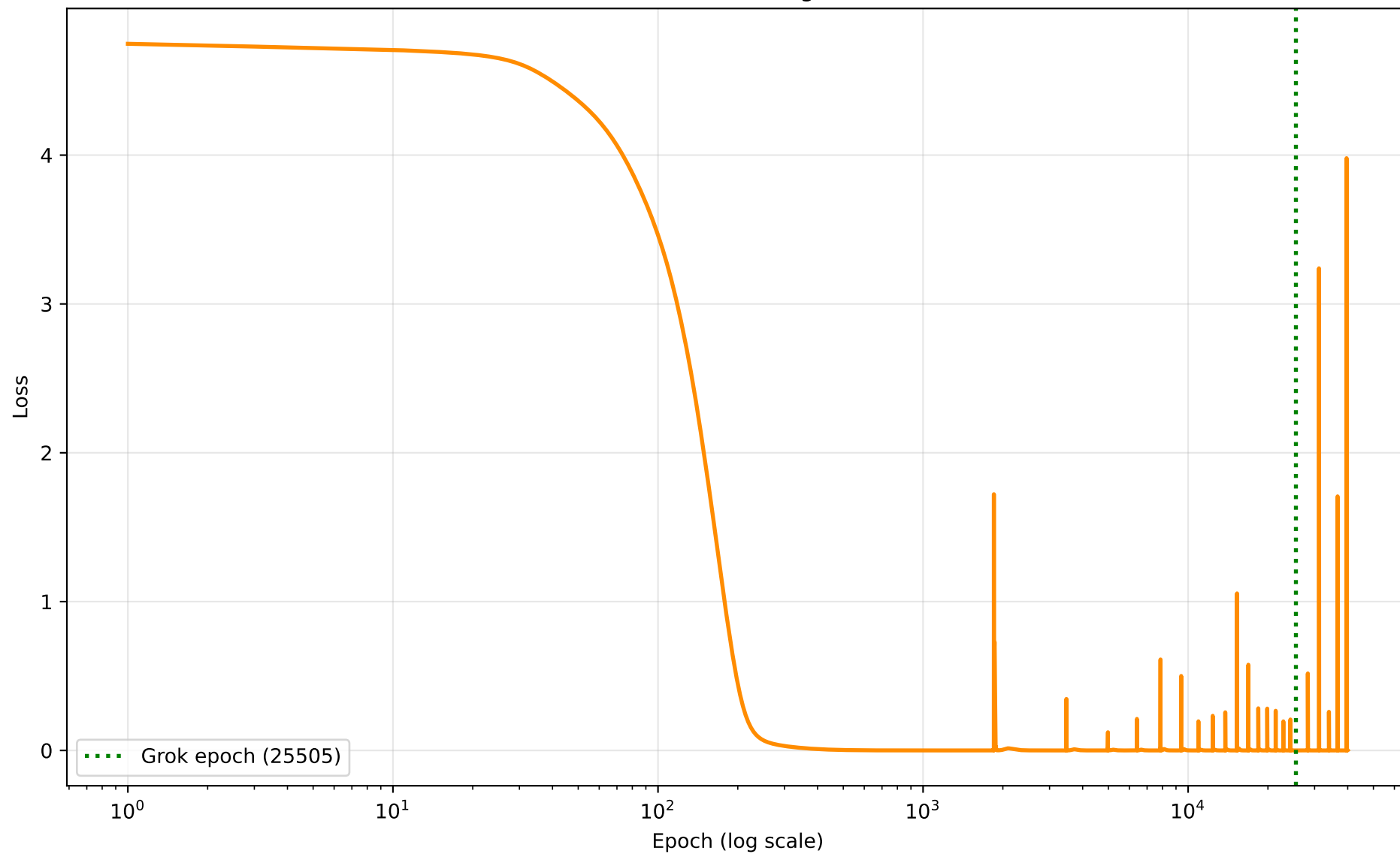
Limit-cycle check : stable

window_start=26005 post_grok_min=0.4869 post_grok_std=0.0099
post_grok_final=1.0000 dips<0.9=20

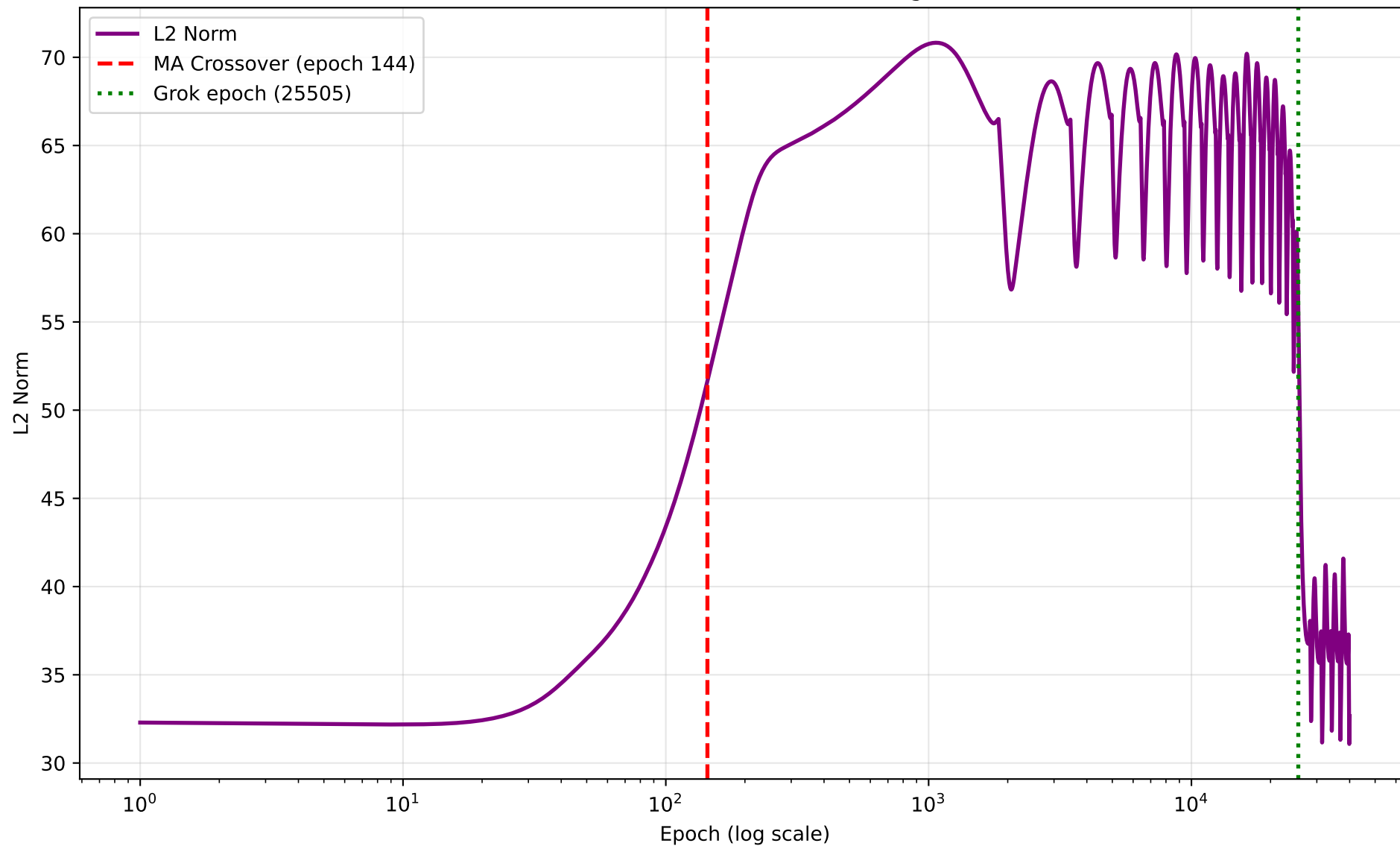
seed 4 — Grokking Curve (grok epoch 25505)



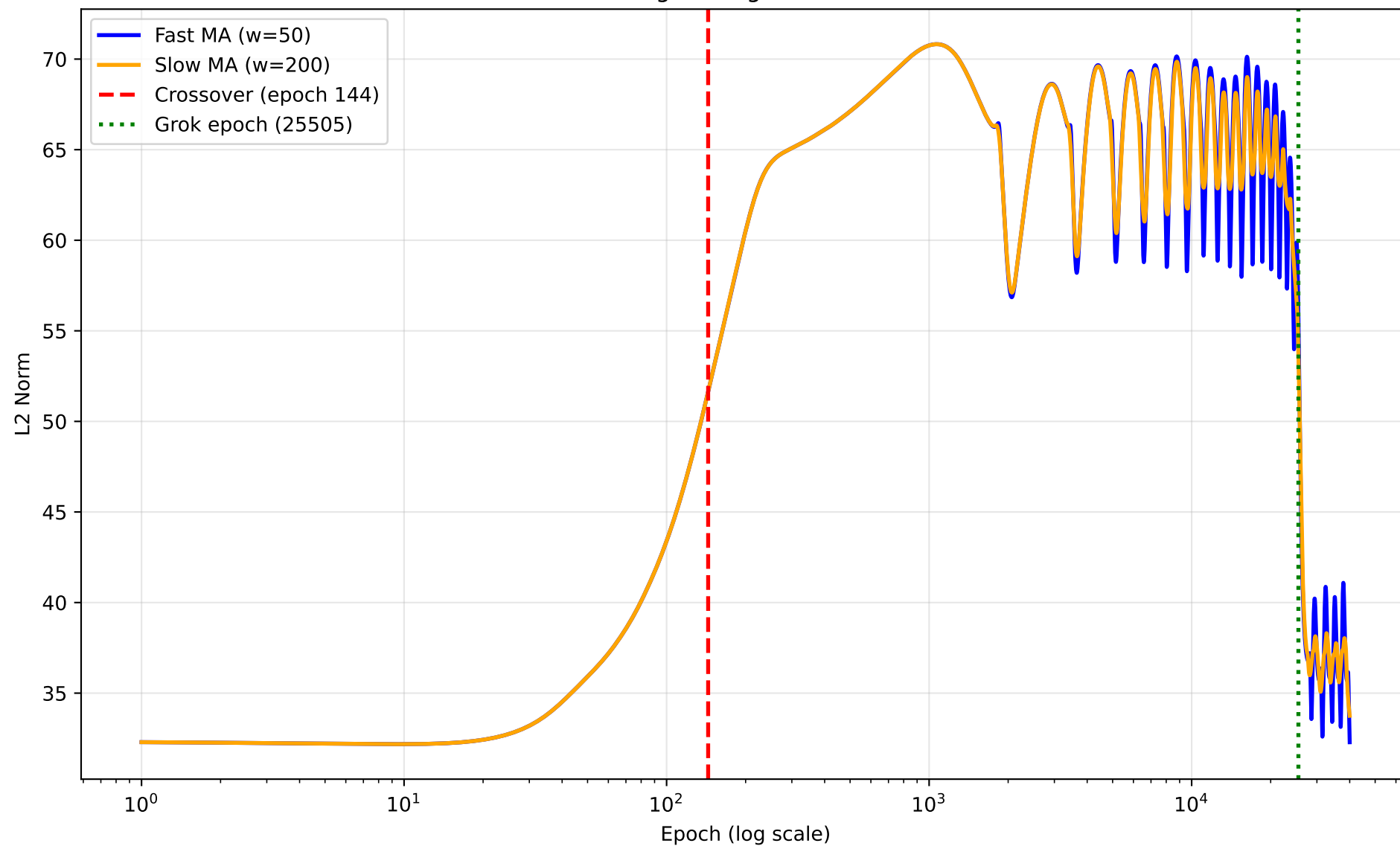
seed 4 — Training Loss



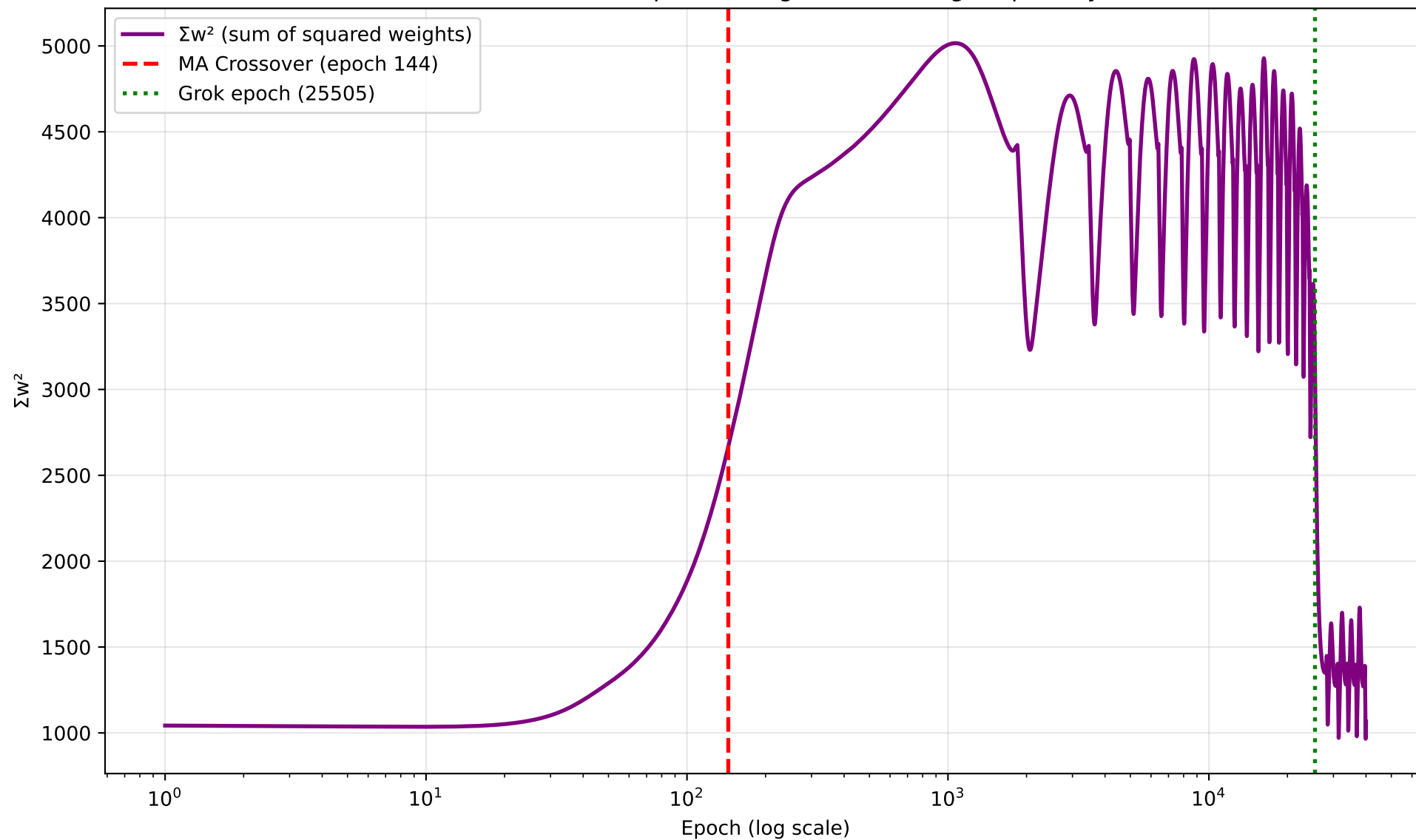
seed 4 — L2 Norm of Model Weights



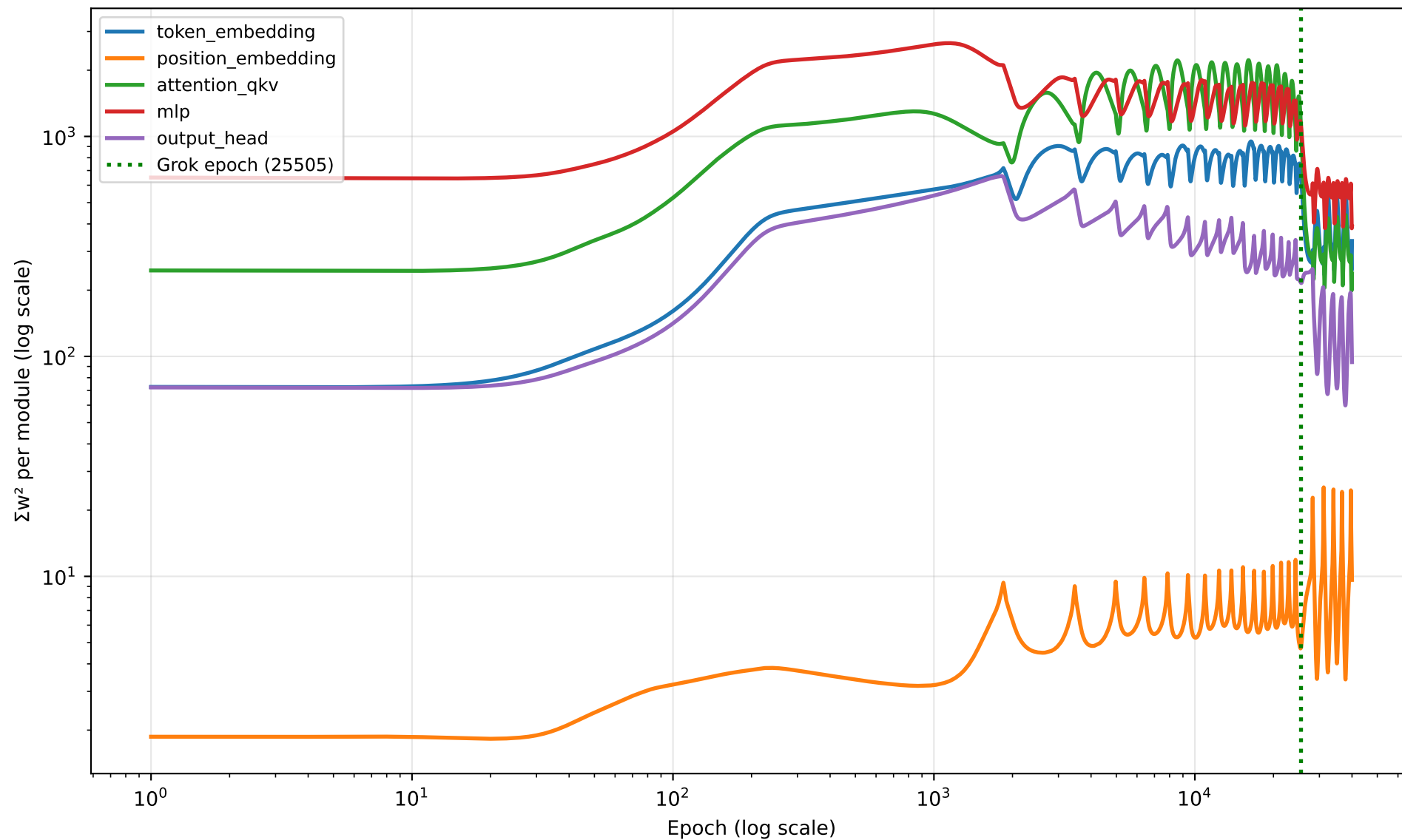
seed 4 — Moving Average Crossover Detection



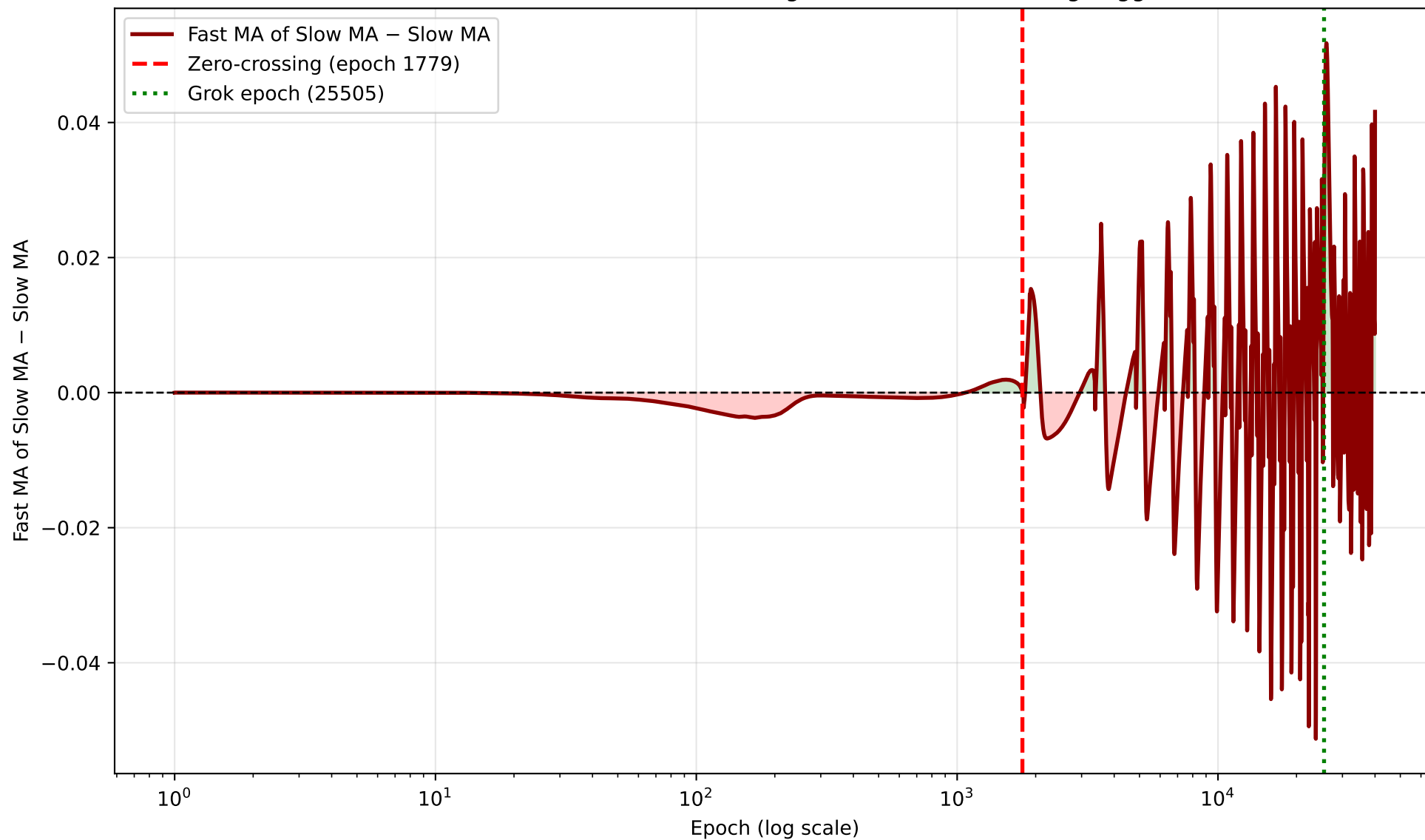
seed 4 — Sum of Squared Weights (Nanda Fig. 7 quantity)



seed 4 — Per-Module Σw^2 Breakdown



seed 4 — MA-of-MA Differential (log scale) — Zero-Crossing Trigger



seed 4 — MA-of-MA Differential (linear scale) — Zero-Crossing Trigger

